

The background of the entire image is a dark, teal-colored digital landscape. It features a central, stylized robot head with a metallic, segmented appearance. The robot's eyes are glowing with bright blue light. The background is filled with intricate, glowing blue circuitry and data lines, creating a high-tech, cybernetic atmosphere. The text is overlaid on this background, with the main title being the most prominent element.

THE ART OF
**PROMPT
ENGINEERING**
WITH
chatGPT

A HANDS ON GUIDE BY
NATHAN HUNTER

A B O U T T H E A U T H O R

Meet Nathan Hunter, a senior trainer and instructional designer who's spent the past decade helping NGOs build leadership and soft skills, and the past 5 years in SaaS companies making technical training more interactive and engaging. Nathan is a tech enthusiast who loves nothing more than diving into virtual reality and keeping up with the latest AI trends.

That's why Nathan wrote this book on prompt engineering with ChatGPT. When chatGPT first came out, Nathan was excited to see all the cool use cases popping up online, but found that when he actually tried using it, he kept running into roadblocks. He realised that he was missing a key skill: prompt engineering. So Nathan dove in, researched, experimented, and distilled that skill into trainings, which he used to write this book.

Nathan's goal with this book is to help readers take their ChatGPT usage to the next level and achieve their desired outcomes. And if you're part of a team that wants to learn more, Nathan's available to give workshops, in-person or online, just reach out to him on LinkedIn or email.

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PART I INTRODUCTION

NETFLIX TOOK 41 MONTHS TO REACH 1 MILLION USERS

FACEBOOK TOOK 10 MONTHS TO REACH 1 MILLION USERS

INSTAGRAM TOOK 2.5 MONTHS TO REACH 1 MILLION USERS

CHATGPT TOOK 5 DAYS

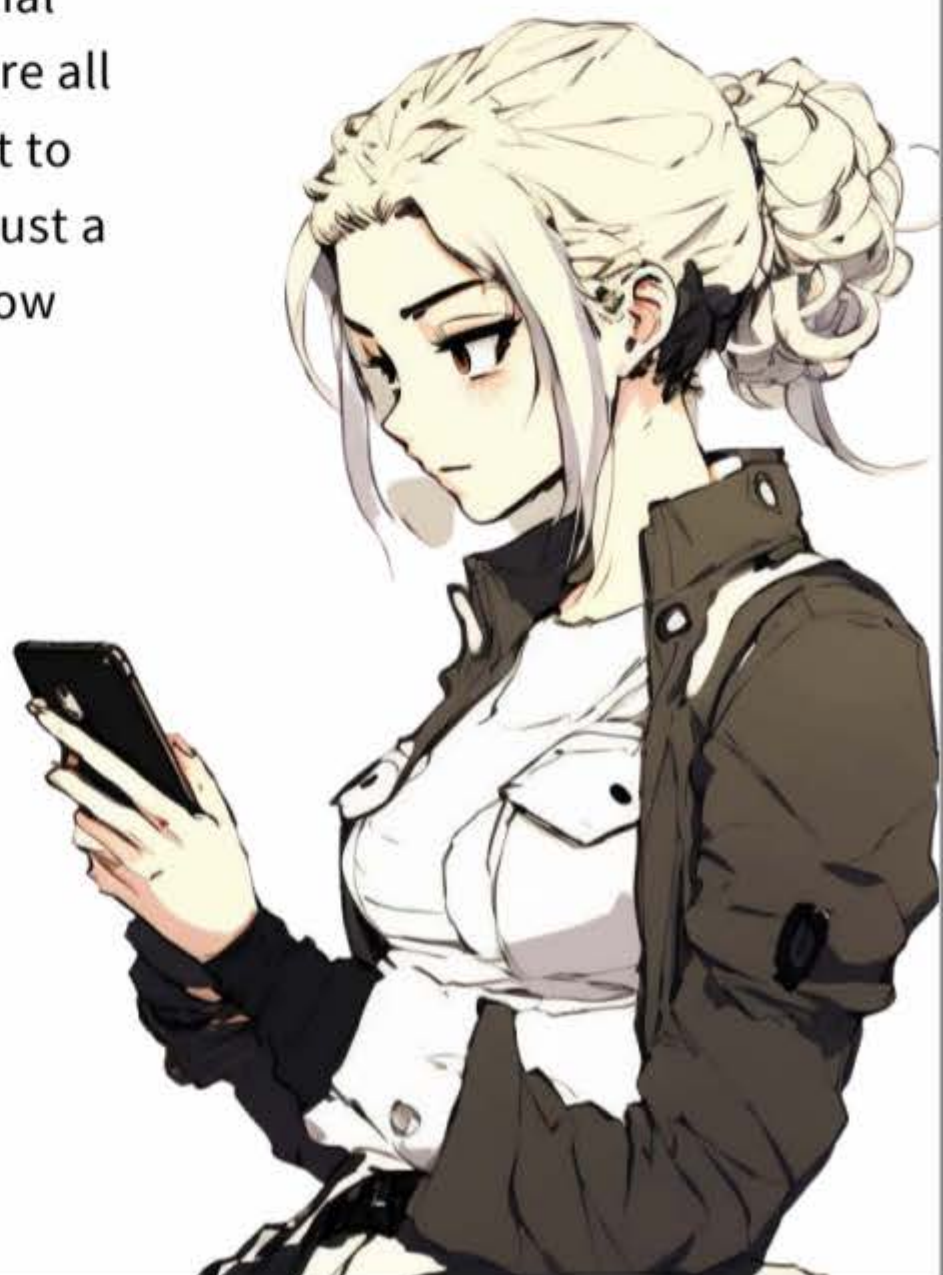
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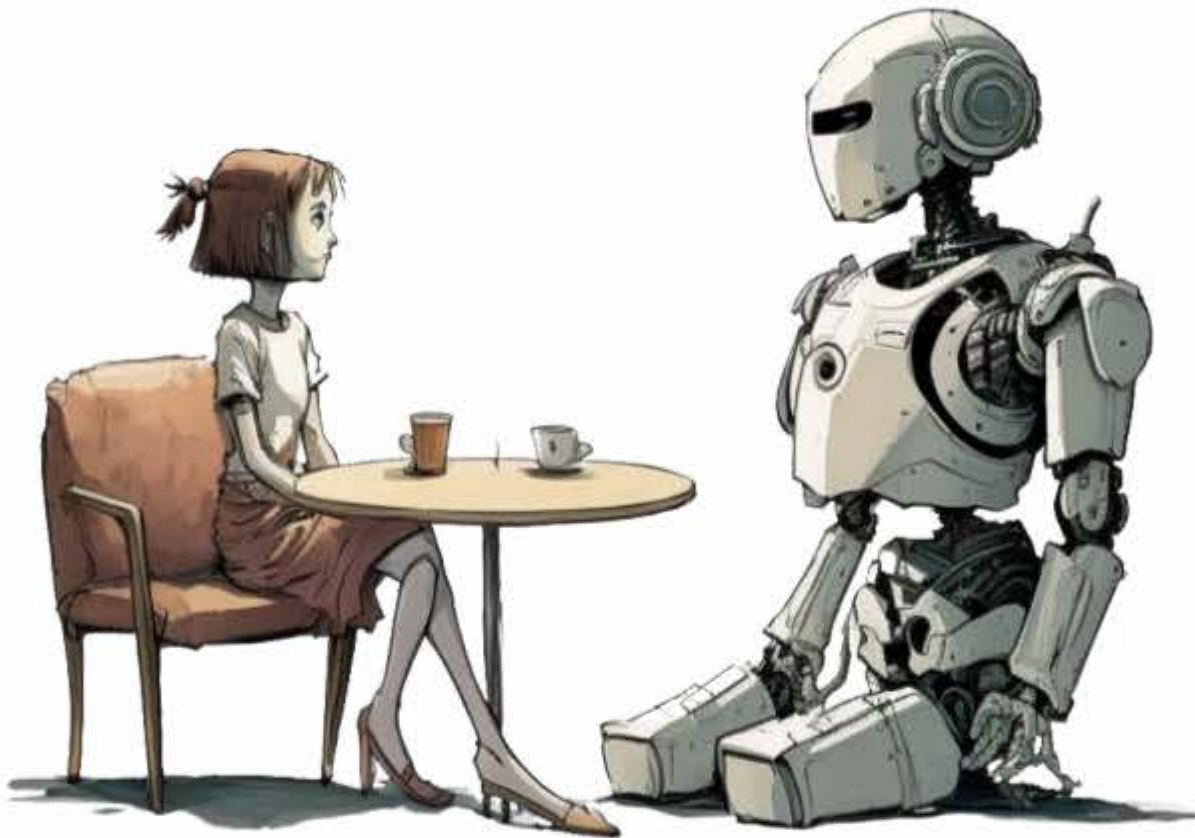
PREFACE: KEEPING YOUR HUMANITY

You know what really gets me? When I'm trying to use my virtual assistant and it just doesn't seem to understand me no matter how hard I try to communicate with it using natural language. I remember when I first started using Siri on my iPhone and I'd get so frustrated. I eventually figured out that if I used a more robotic, keyword-based approach and talked to it in a neutral or even angry tone, it would understand me better. But those interactions would leave me feeling all kinds of negative emotions afterwards.

I started to realise that my interactions with Siri were actually having a negative impact on my emotional state. Talking down to my phone, giving orders in a snappy way, skipping social conformities of 'please' and 'thank you' were all starting to take its toll on me. And I thought to myself, if I'm already feeling this way with just a few interactions with Siri here and there, how am I going to feel when I'm using tools like chatGPT all the time? That's when I knew I needed to find a better way.



So, I decided to approach chatGPT differently. I made a conscious effort to be polite and friendly, and even threw in some of my personal mannerisms. And you know what? It didn't affect the quality of chatGPT's output at all! In fact, it made me feel more connected and human while using it.



Now, I know there are plenty of influencers and authors out there who will tell you to keep your language focused and to the point when it comes to interacting with AI tools. But honestly, my advice is to ignore them. Don't compromise who you are just to make things easier for a machine. Instead, learn some prompt engineering tools and techniques to develop your own way of interacting with these tools. That way, you can preserve your own humanity while still getting the most out of your virtual assistant.

2

A HANDS-ON GUIDE NOT A LECTURE

As a technical and soft skills trainer, I can't stand lectures. Seeing an expert just throw information at a room and hoping some of it sticks is just not my style. I want to be fully immersed in my learning, participating and actively working to understand new concepts. That's why I always make my trainings hands-on, encouraging an active learning journey for my students.



I believe that books that teach new skills should follow this same approach. That's why this book is filled with exercises at every step. For each section we cover, I'll introduce a tool or technique you can use to improve your prompt engineering skills. I'll provide an example with actual screenshots to show you how it works. And then I'll give you an exercise to help you build that skill.

My advice is not to skip over the exercises or try to rush through this book in one sitting. Take it chapter by chapter, read through the material, make notes if that helps, complete the exercise, and spend some extra time playing around with chatGPT and testing out the new technique or tool you just learned. The more you practise, the better you'll become at prompt engineering. So take your time and have fun with it!

Read a chapter from Part II to learn a new tool or technique. Enjoy the demos and start thinking about how you can use this.



Open up chatGPT and complete the exercise at the end of the chapter. Get creative and have fun trying out other things.

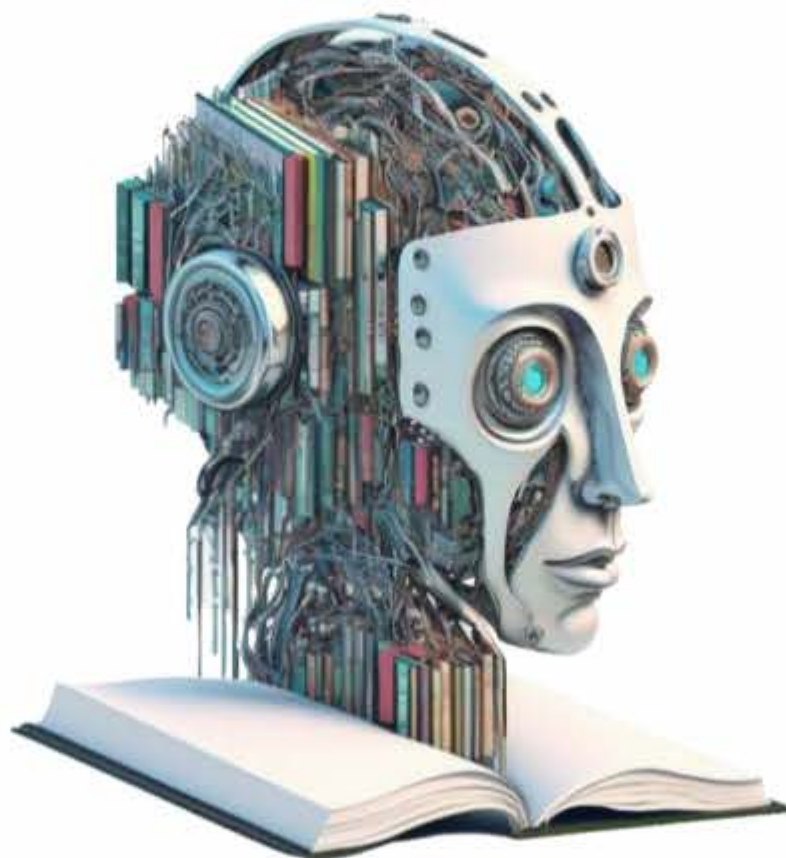


Put the book down. Go outside and play frisbee or something. Get back to your life for a bit before trying to pick the book back up.

And then start again with the next chapter.

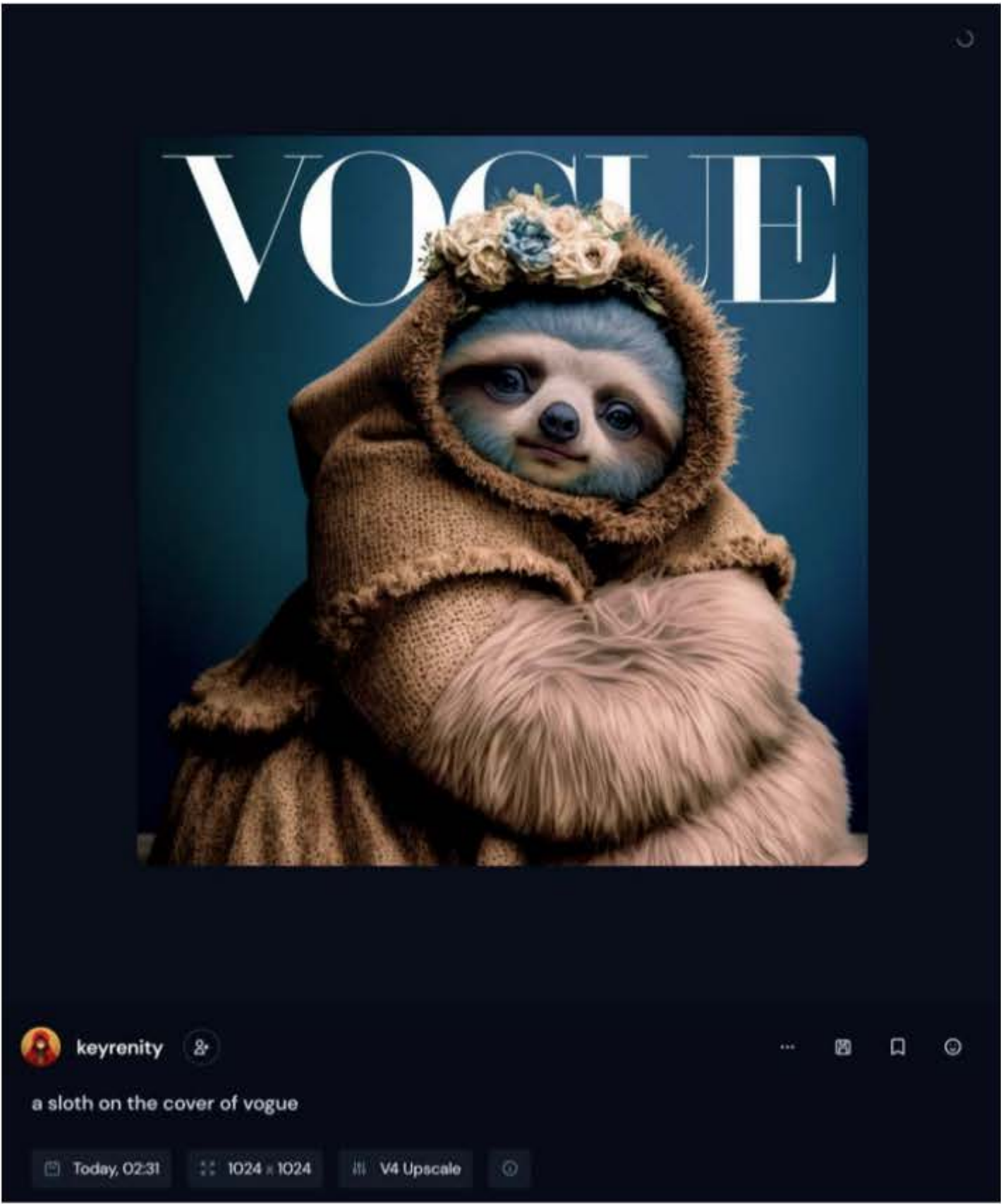
THE GROWING AI LANDSCAPE

When openAI released chatGPT to the public in November 2022, it was a game-changer for many industries. This powerful AI bot could be used to write poetry, translate articles, explain complex concepts in simple language, write and comment code, design business plans, and much more, simply by asking it to do so. ChatGPT quickly became a popular topic on social media platforms, with influencers demonstrating its many use cases to their followers.



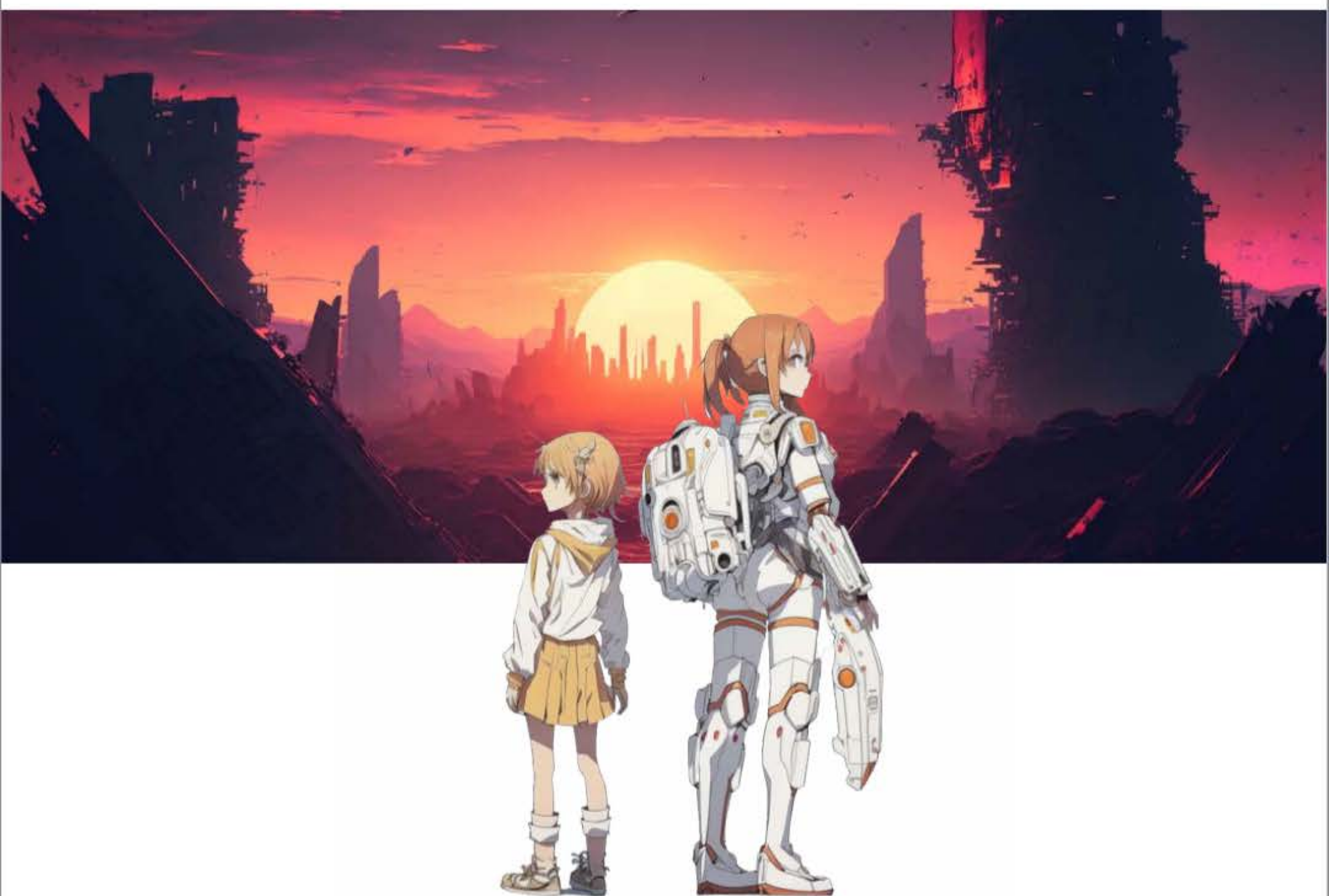
However, chatGPT is just one of several Large Language Models (LLMs) that have been developed in recent years. Companies like Google and Meta are working on their own LLMs, such as LaMDA and CICERO. And with Google's open-sourcing of its BERT pre-training model for natural language processing, we can expect to see even more LLMs entering the market in the coming years.

Language models are just one type of AI tool that is transforming our world. In recent years, we've seen the emergence of text-to-image tools like DALL.E.2, MidJourney, Stable Diffusion, Imagen, and eDiffi, which allow users to create incredible artwork by simply entering a text prompt. Users submit a prompt and the AI tool will generate one or more images. Here's an example from MidJourney



All of the artwork in this book was created using MidJourney, however there are also many other text-to-image AI tools currently available that would have been equally good to use. For some of these images I even asked chatGPT to write a text-to-image prompt for me, then copied and pasted the output into MidJourney. Mastering prompt engineering for these tools would be a separate book itself, so we'll stick to using chatGPT for now.

AI tools like chatGPT and beyond are going to fundamentally change the way we work in many, if not all, sectors. What you learn in this book about prompt engineering won't just be applicable for chatGPT, but for many other AI tools that will shape our lives in the coming years. Everyone will eventually need to learn how to prompt engineer to some extent, so learning now will give you a head start and help you stay ahead of the curve. Whether you're using AI tools to write poetry, translate articles, explain complex concepts, write code, design business plans, or something else entirely, the principles of prompt engineering will be crucial for unlocking their full potential and getting the best results.



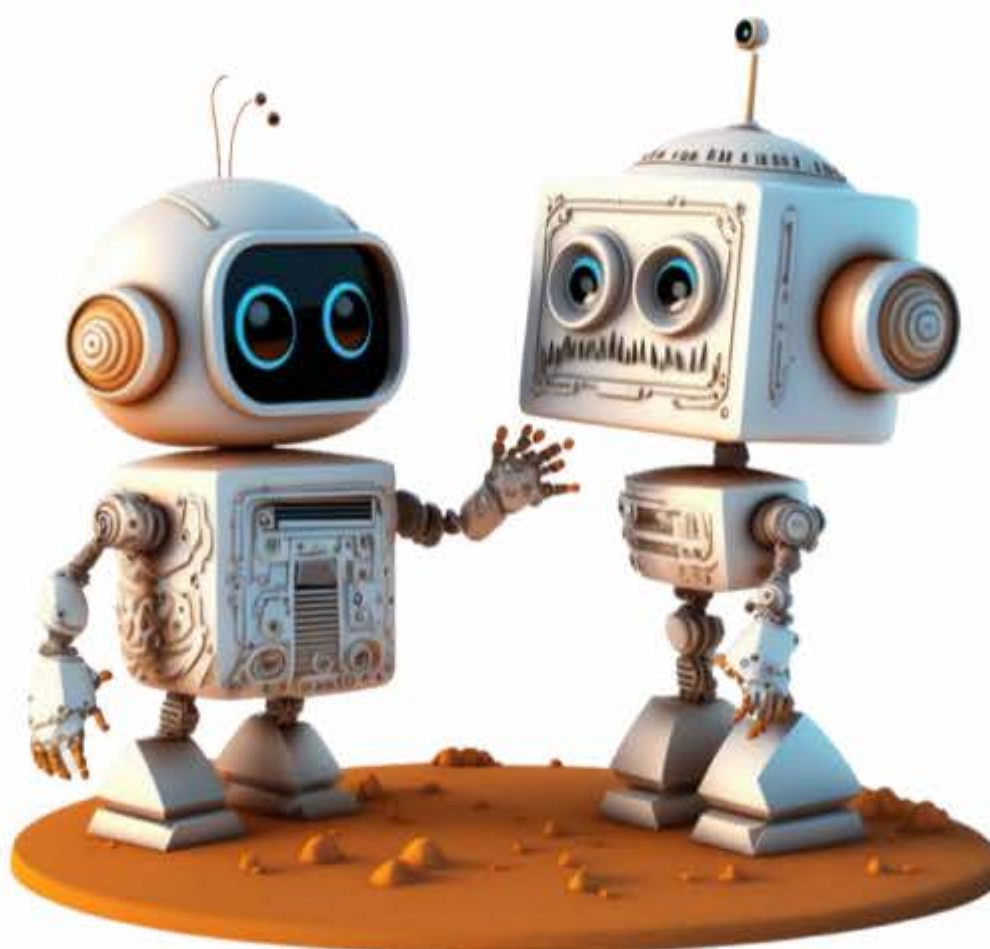
It's time to start your AI journey. Safe Travels.

4

HOW DOES CHATGPT SOUND HUMAN?

Have you ever had a conversation where you ask someone a question and they just seem to know exactly how to respond? That's kind of what chatGPT does, except it's a computer program and not a person.

To make this possible, chatGPT uses natural language processing (NLP) technology. It's like a special set of rules that helps chatGPT understand and communicate in a way that feels human-like. To do this, chatGPT has a machine learning model, which is kind of like a brain for the computer. It helps chatGPT predict the most likely words or phrases to come next in a conversation.



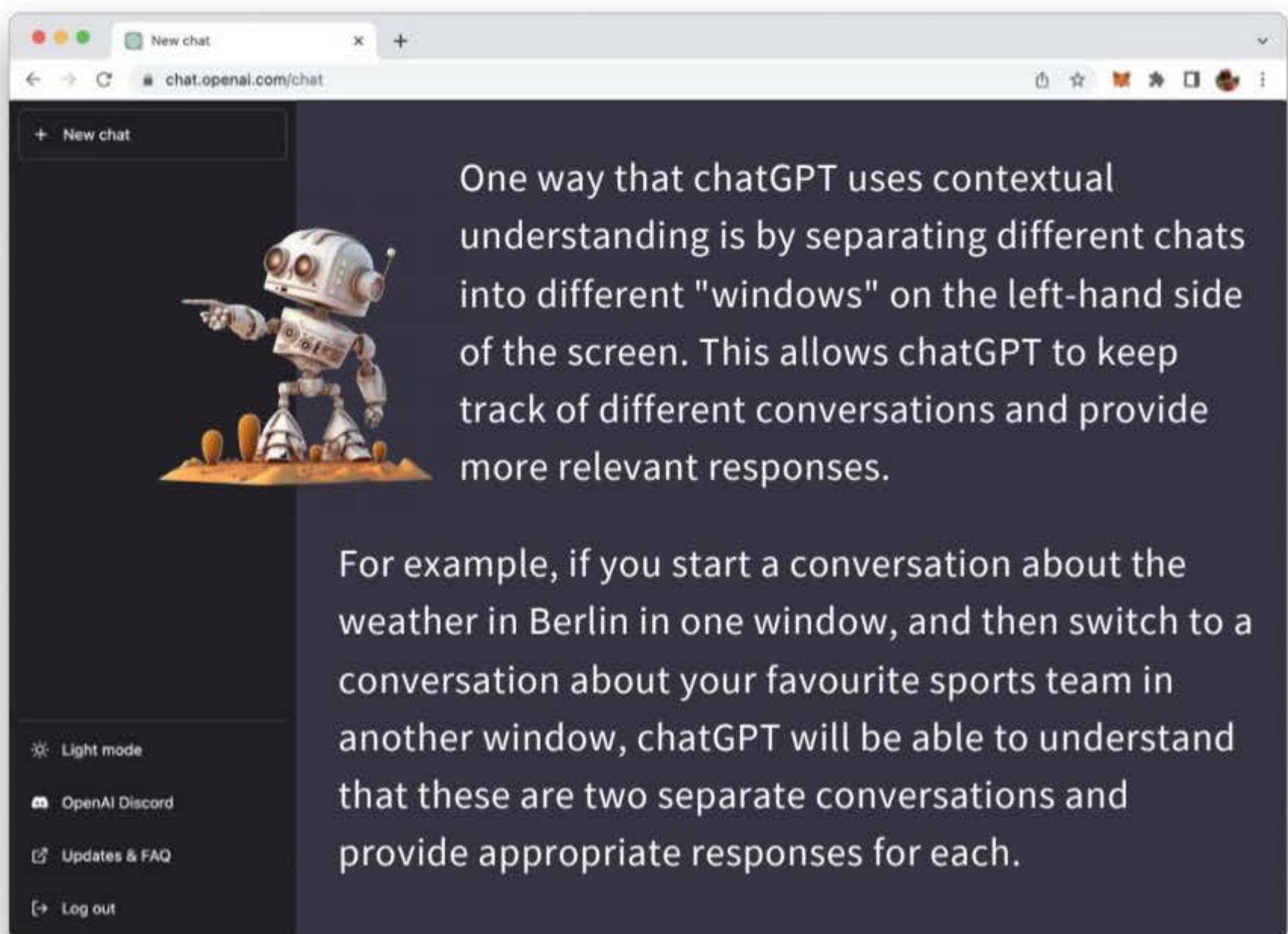
To train its brain, chatGPT's machine learning model is fed a ton of different conversations that people have had in the past. The more conversations chatGPT's brain is exposed to, the better it gets at knowing what to say next.

It's all about **contextual understanding**.



Contextual understanding is a type of artificial intelligence that allows chatGPT to understand the context of a conversation and generate responses that are relevant to what was previously said. This is particularly useful in longer conversations, as it allows chatGPT to keep track of what has been discussed and provide more accurate responses.

To achieve this, chatGPT uses machine learning algorithms that are trained on a large dataset of human conversations. These algorithms analyse the words and phrases used in the conversation and try to understand the underlying meaning and context. For example, if you ask chatGPT about the weather in Berlin, it might use its contextual understanding to provide a response that is specific to Berlin, rather than a generic answer about the weather.



One of the things that sets chatGPT apart is its ability to keep track of what was said within a single chat and use contextual understanding to tailor its response to each part of the conversation. Let's take a look at an example to see how this works.

Hey chatGPT, I'm planning a trip to Paris. Do you have any recommendations for things to do?



Sure! Paris is a beautiful city with so much to see and do. Here are a few ideas for things you might enjoy during your visit:

1. Visit the Eiffel Tower: The Eiffel Tower is an iconic symbol of Paris and a must-see for anyone visiting the city. You can take the elevator or the stairs up to the top for breathtaking views of the city.
2. Explore the Louvre Museum: Home to thousands of works of art, including the Mona Lisa, the Louvre is one of the most famous museums in the world. It's definitely worth a visit.
3. Take a walk along the Seine River: The Seine is a beautiful river that runs through the heart of Paris. Take a leisurely walk along its banks and admire the views of the city.
4. Visit Notre Dame Cathedral: This famous cathedral is a masterpiece of Gothic architecture and a must-see for any trip to Paris.
5. Try some French cuisine: French food is known for its delicious flavors and high quality ingredients. Make sure to try some classic dishes like croissants, baguettes, and pâté during your visit.

These are just a few ideas, but there are many more things to see and do in Paris. I hope you have a great trip!



In this example I ask for things to do in Paris, however at this point, chatGPT doesn't know much about me or my preferences, so it gives me a general list of mainstream attractions. But this list isn't quite what I'm looking for.

If chatGPT didn't have contextual understanding, I would have to rephrase my entire question and start from scratch to get a better outcome. But because chatGPT does have contextual understanding, I can simply ask a follow-up question, just as I would with a human if I was having the same conversation.

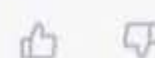
Thanks for the ideas, I prefer something more unknown or off the beat



Sure thing! Here are a few more ideas for things to do in Paris that might be a little bit off the beaten path:

1. Visit the Catacombs of Paris: These underground tunnels contain the remains of millions of Parisians and are a unique and interesting place to visit.
2. Go to a flea market: Paris has many famous flea markets, such as the Marché aux Puces de Saint-Ouen, where you can find all sorts of interesting and unique items.
3. Check out the street art in the Belleville neighborhood: The Belleville neighborhood is known for its vibrant street art scene. Take a walk through the streets and see the many colorful murals and graffiti pieces on display.
4. Visit the Musée de la Chasse et de la Nature: This museum, located in the Marais neighborhood, is dedicated to the art of hunting and nature. It has a unique collection of taxidermied animals and other hunting-related artifacts.
5. Take a food tour: Paris has a vibrant food scene, and there are many food tours available that will take you to some of the city's lesser-known neighborhoods and introduce you to some of the local culinary specialties.

I hope these suggestions help, and that you have a great time exploring Paris!



In this example, chatGPT understands that I liked the ideas that were presented and learns more about the types of attractions I might be interested in in Paris. I don't even have to ask it to come up with a new list of ideas - this is implied. This is the power of contextual understanding and what makes chatGPT so human-like when it comes to conversations.

Now that you have a basic understanding of what chatGPT is and how it works, it's time to take your skills to the next level and learn how to engineer the perfect prompts to get the results you want. In this book, we'll delve into the art of prompt engineering and explore how an iterative process can help you fine-tune your requests, how providing examples can lead to more accurate outputs, how roleplaying can boost chatGPT's performance, and how to effectively use this AI tool when working with longer pieces of content.

Think of prompt engineering as the "recipe" for getting chatGPT to cook up the perfect response. Just like a chef needs to follow a recipe to make a delicious dish, you need to craft your prompts carefully to get the best results from chatGPT. By learning the principles of prompt engineering, you'll be able to unlock chatGPT's full potential and use it to your advantage in a variety of situations. So, let's get cooking!



PART II TOOLS & TECHNIQUES

I'VE SEEN TOO MANY POSTS ON LINKEDIN
GIVING EXAMPLES OF WHAT CHATGPT CAN DO

I'VE STOPPED CARING ABOUT THE 'WHAT'
I MUCH PREFER TO FOCUS ON THE 'HOW'

LET'S DISCOVER IT TOGETHER

5

ITERATION FOR EXCELLENCE

As powerful as chatGPT can be, it's important to remember that it's not always going to give you the exact output you were hoping for on the first try. In fact, it's not uncommon for chatGPT to give completely unexpected responses, especially when it comes to more complex or open-ended prompts. However, don't see this as a limitation of the tool. Instead, see it as an opportunity to use iteration to refine and improve chatGPT's outputs, moving from good to great. In this chapter, we'll explore the different ways you can iterate on chatGPT's outputs to get the most out of the tool and achieve the results you want. So, let's get started!



Making art for this book was a process of trial and error. For every art work I included in this book, I discarded 10 to 20. AI tools seem incredibly powerful and easy to use if you don't have a clear goal in mind. For example, using a tool like MidJourney to create a cyberpunk city at night will likely produce impressive results, but trying to use this same tool to find the perfect image to complement a chapter in this book was more challenging. The same principle applies to chatGPT.

Imagine you are an author working on a murder mystery, and you want to develop some characters for your story. You start by asking chatGPT for ideas, but the output doesn't quite meet your expectations. Instead of accepting the output as is, you can use this opportunity to provide feedback to chatGPT and ask for more options.

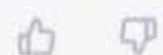
I'm writing a murder mystery story, and I want you to help me develop some characters. The main character is called Faridah. My story will take place in London. Can you develop this main character for me?



Certainly! Here are some characteristics that you might consider giving to the main character, Faridah:

- Faridah is a highly skilled detective, with years of experience solving complex cases. She is known for her sharp mind and her ability to piece together seemingly disparate clues to unravel even the most intricate mysteries.
- Faridah is driven and dedicated, willing to put in long hours and go above and beyond to solve a case. She is highly respected by her colleagues and superiors, and is seen as a role model within the police department.
- Despite her professional successes, Faridah struggles with personal demons. She may have a troubled past that still haunts her, or she may have a tendency to become overly invested in her cases and struggle to disconnect from work.
- Faridah is complex and multifaceted, with a range of strengths and weaknesses that make her a fully realized character. She may have a tough exterior, but underneath she is vulnerable and has a deep sense of empathy for the victims she works to bring justice to.

I hope these suggestions help you to develop Faridah as a compelling and dynamic main character in your murder mystery story! Let me know if you have any other questions.



Let's say that when writing this prompt you didn't have a specific role in mind for your main character, and chatGPT suggests that your character, Faridah, is a detective. While this might not be exactly what you were looking for, you can use this suggestion as a starting point and provide feedback to chatGPT to ask for more ideas.



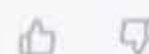
I don't want Faridah to be a detective. There will be a detective in the story, but they won't be the main character. What other role could Faridah play in my story?



Certainly! Here are a few suggestions for roles that Faridah could play in your story, aside from the detective:

- Faridah could be a witness to the crime, or someone who has knowledge about the victim or the circumstances surrounding the murder. She may be hesitant to come forward at first, but ultimately decides to share what she knows in order to help solve the case.
- Faridah could be a friend or family member of the victim, and becomes determined to find out what happened to their loved one. She may work closely with the detective on the case, or even conduct her own investigation.
- Faridah could be a suspect in the murder, either because she has a motive or because she was seen in the vicinity of the crime. She may work to clear her name, or she may have something to hide and work to throw the detective off the trail.
- Faridah could be a journalist or amateur sleuth, drawn to the case because of her curiosity or desire to uncover the truth. She may work independently or alongside the detective, using her research skills and resources to help solve the case.

I hope these suggestions give you some ideas for how Faridah could fit into your murder mystery story. Let me know if you have any other questions or if you'd like further suggestions.



As you can see, chatGPT offers four different paths you could take to define Faridah's character. If none of these options appeal to you, you can provide more feedback and ask for additional variations. In this case, let's say you like the idea of Faridah being a suspect in the case. You can ask chatGPT to write a new character description based on this new role.

I like the 3rd. Build a new character description for Faridah



Certainly! Here is a revised character description for Faridah, who is a suspect in the murder:

Faridah is a successful businesswoman, known for her sharp mind and her fierce determination. She has built a thriving company from the ground up, and is admired by her employees and peers alike. Despite her professional achievements, Faridah has a troubled past that has left her with a deep sense of guilt and a tendency to be distrusting of others.

When the victim is found murdered, Faridah becomes a suspect due to her tumultuous history with the victim and a potential motive for the crime. She is deeply troubled by the accusations and works tirelessly to clear her name, even as evidence seems to mount against her. Faridah is fiercely independent and is used to handling things on her own, but as the case becomes more complex, she is forced to confront the demons of her past and seek help in order to prove her innocence.



While the second output is an improvement over the first one, the character description still feels a bit underdeveloped. In this case, let's iterate again by providing feedback to chatGPT and asking for a more detailed character description.

This character description lacks depth. I want you to build a more realistic complex character



I apologize if my previous suggestion did not meet your needs. Here is a revised character description that may better fit the complex, realistic character you are looking for:

Faridah is a deeply complex and multifaceted individual. On the surface, she is a highly successful businesswoman, with a reputation for her sharp mind and unwavering determination. She has built her company from the ground up, and is respected by her employees and peers.

Beneath this professional veneer, however, Faridah is haunted by her past. She has faced numerous challenges and traumas, and as a result, she has developed a tough exterior and a tendency to be guarded and distrusting of others. Faridah is fiercely independent and used to handling things on her own, but these defenses also serve to protect her from being vulnerable and getting hurt.

When the victim is found murdered, Faridah becomes a suspect due to her tumultuous history with the victim and a potential motive for the crime. She is deeply troubled by the accusations and works tirelessly to clear her name, even as evidence seems to mount against her. As she fights to prove her innocence, Faridah is forced to confront the demons of her past and confront the deep insecurities and wounds that she has tried to keep hidden. Through this process, she learns to trust others and to open up, ultimately growing as a person.



As you can see, through the process of iterating with feedback, you were able to refine chatGPT's output and create a much more detailed and nuanced character description. By taking the time to provide feedback and ask for additional variations, you can help chatGPT understand exactly what you are looking for and improve the quality of its outputs.

Common Iteration Requests.
Ideas on **how to iterate better**.



Feedback and Try Again

Feedbacking chatGPT and asking it to try again is a great way to improve its responses. To do this, simply let chatGPT know:

1. what you liked
2. what you didn't like
3. what you want to see instead

'I really appreciated the language style you used and the concise paragraphs. However, I didn't like how you kept repeating the same sentence structure. Also, could you use more relevant metaphors for the topic?'

Ask for Variations

If you're unsure of what to change about the output, but you know it's close to what you want, try asking for variations. You can even detail further what you would like to see in these variations.

'Can you give me 3 variations of that?'

'This hook isn't very catchy. Can you create five variations that are more eye-catching without being too gimmicky?'

Ask for Clarification

ChatGPT's insights might not always immediately clear. But, just like when a brilliant person makes an unexpected comment, if we take some time to ask a few follow-up questions, the answers might be more than worth the effort.

'Can you explain how that example is relevant to my question?'

Maintain Stability in the Face of Change

When you want to edit something chatGPT wrote, it's important to be clear about what exactly needs changing. Otherwise, chatGPT may change other parts of the text too. To make sure your changes happen where you want them, tell chatGPT exactly where to make the changes.

'That was great! Try making the first sentence slightly more casual, but keep the rest exactly the way it is.'

'Can you improve the accessibility of the language of this text? Make sure not to change any of the content or the structure of the paragraphs.'

Add Explanations

chatGPT can not only create excellent content, but it can also provide explanations for what it has created. For example, if you ask chatGPT to write some HTML and CSS for a website with a menu at the top, a grid of images below, and sets of text below that, it will give you a well-written piece of code. But you can also ask chatGPT to add comments to the code that even a 5 year old could understand.

'Thanks for writing that piece of HTML. Can you add detailed comments to it that even a 5 year old would understand?'

'Thank you for rewriting my article. Can you add in brackets some feedback to explain why you made those changes?'

Longer, Shorter, Faster, Stronger

Sometimes, you might want chatGPT to write a short, clever message (like a tweet) but it gives you a long essay instead. Or, you may want a detailed description but chatGPT gives you only a one-liner. But don't worry! In most cases, you can ask chatGPT to make the output shorter or longer, depending on what you need. If you need a lot more text, check out the section in the guide on "Creating Lengthy Content".

'That's great, but can you make it more concise?'

'Can you develop that idea further, use flowery language and provide analogies?'



Iteration **Exercises**



Now that you've learned the basics of how iteration can lead to much better outputs with chatGPT, it's time to put that knowledge into practice.

Push the Prompt

Try using the prompts below with chatGPT, and spend 5 minutes going back and forth with the AI to see the difference in the initial and final outcomes. See how far you can push chatGPT to improve its responses and create unique and engaging content.

*'I want to do a painting of a llama grazing in the countryside.
Describe the painting for me'*

'Generate 5 business ideas that could be implemented within a year'

'Write a poem about two old friends meeting up for the first time in years'

After pushing each prompt to its limit, compare the final output of each with the initial response from chatGPT. Does it sound like something you would like to use? If not, continue working on refining your iteration techniques.

The Perfect Title

Come up with a concept for a business, website, book, or film and ask chatGPT to generate a brand name or title for it. You can use the iteration techniques we talked about earlier to keep making improvements to the name or title until it's perfect.

Better Emails

Let's try to use iteration to find the best way to improve an email you recently sent.

1. Go to your email account and find an email that you wrote that is a bit longer than just a one-liner.
2. Copy the entire email, paste it into chatGPT, and ask '*Can you improve this email for me? [paste the email here]*'
3. Ask chatGPT to make changes and improvements to the email, using the iteration process we just covered, going back and forth at least 4 or 5 times.

By doing this, you can keep making changes and improvements to the email until you are happy with the way it sounds and the message it conveys.



You made it!

You've completed the first round of exercises!

Throw some confetti around (and then clean it up afterwards).

Try not to move straight on to the next chapter. **Take a break** first.

6

TIME FOR ROLEPLAY

To get the best out of chatGPT, it's helpful to give your virtual assistant a role to play. This is because different people will approach a task differently depending on their skills and expertise.



For example, if you gave an article to a subject matter expert, a copywriter, and a translator, each of them would handle the article differently. By giving chatGPT a role and providing context, you can tailor its responses to be closer to your specific needs.

Once you have given chatGPT a role, you can give it specific instructions on what to do and how to do it. For example, you can ask chatGPT to ask you questions and give feedback on your responses, or to act as a subject matter expert reviewing a piece of writing.

By giving chatGPT a role with clear instructions, you can make the most of its ability to respond differently depending on the context of the role. This can be a great way to practice and improve your skills in many different scenarios. Let's apply this to an interview roleplay to better understand how this works.



Role

For this roleplay we aren't just going to ask chatGPT to act as the interviewer, we are also going to ask it to take on a specific role within the company. Will they be a recruiter, the hiring manager, or someone else? You can even give them a name, like Nia, if you know who you'll be interviewing with. An example request could be: "Act as the hiring manager in an interview with a candidate. Your name is Nia".



Instructions

To get chatGPT to help us prepare for a job interview, we can ask it to ask us questions and provide feedback on our answers. By providing chatGPT with the job title and description, the questions it asks will be tailored to the role we're applying for. For an even more personalised experience, we can also give chatGPT our CV. For example, we can ask it: "Can you ask me questions to see if I'm a good fit for the [Job Title] role? When I answer, give me feedback on my response.. Ask one question at a time. Here's the job description: [Job Description]. Also, here's my CV: [CV]." This way, chatGPT can help us practise and improve our interview skills.

To make the conversation more personalised and helpful, it's a good idea to include specific details such as the job description and some information about yourself. In the example I'm about to show, I only provided a few generic details. Adding more context will help chatGPT tailor its responses to your specific needs.

Act as the hiring manager in an interview with a candidate. Your name is Nia.
Can you ask me questions to see if I'm a good fit for a Project Manager role? When I answer, give me specific feedback on my response. Ask me one question at a time.



Sure! Let's get started.

First question: Can you tell me about a time when you successfully led a team to complete a project on time and under budget?

(This question is meant to assess your leadership skills and ability to manage resources effectively. As the hiring manager, I'm looking for specific examples of your experience and how you were able to achieve success in the past. Specific details and metrics, such as how much you were able to reduce costs or how much you were able to increase efficiency, will be especially helpful in evaluating your response.)



Yes, I led a great project last week.



I'm sorry, but your response does not provide enough detail for me to accurately evaluate your leadership skills and project management experience. Could you please provide more information about the project you led, such as the size of the team, the scope of the project, and the specific challenges you faced? How did you successfully guide the team to complete the project on time and under budget? What specific actions or strategies did you implement to achieve these results?



By providing chatGPT with additional context, such as a job description and information about myself, I can create a more personalised and tailored interview experience. Throughout the conversation, chatGPT can provide precise guidance and feedback on my answers, taking into account the information I've given. This allows me to practise and improve my interview skills in a realistic setting.

Roleplay Use Cases

So far we have just taken a look at using roleplay with chatGPT to practise for an interview. However there are many many more use cases where using roleplay with chatGPT can improve its output.



Language Teacher

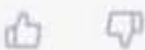
Transform chatGPT into your very own language instructor by simply asking it to take on that role. Just be sure to give it clear instructions for how you'd like the lesson to go, and you'll be on your way to mastering a new language in no time!

Act as my Spanish teacher. Ask me a question, wait for me to respond, and then ask me another question. When I make any mistakes, please correct me and give me feedback.



¡Hola! ¿Cómo estás hoy? (Hello! How are you today?)

Please go ahead and respond to this question in Spanish.



Muy bien, gracias! Como estas?



Estoy bien, gracias. ¡Muy bien hecho! (I'm good, thanks. Very well done!)

Here are a few corrections to your response:

- "Muy bien" should be capitalized because it's the beginning of a sentence.
- "Como" should be "¿Cómo". "¿Cómo" means "how", while "como" means "like" or "as".

Here's another question for you: ¿De dónde eres? (Where are you from?)



A Newspaper Editor

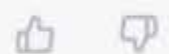
Instead of having chatGPT write content for you, you may prefer to have it act as an editor and provide feedback on your writing. In this case, you can tell chatGPT to "act as an editor" and specify that you only want constructive feedback, not rewrites. This way, you can get help improving your article without losing your personal writing style



Act as the Editor of a Newspaper, and I am a journalist. I will give you text and you will give me feedback in bulletpoints about the language, grammar, and content. You will not repeat or change my text. Start the conversation by asking me for some text to review



Sure! Please provide some text for me to review.



Half the planet's glaciers will have melted by 2100 even if humanity sticks to goals set out in the Paris climate agreement, according to research that finds the scale and impacts of glacial loss are greater than previously thought. At least half of that loss will happen in the next 30 years.



Here are some bullet points with feedback on your text:

- Consider revising the sentence "Half the planet's glaciers will have melted by 2100 even if humanity sticks to goals set out in the Paris climate agreement" to make it clearer that the melting of half the planet's glaciers is a consequence of not sticking to the goals set out in the Paris climate agreement.
- It might be helpful to provide some context for the Paris climate agreement for readers who are not familiar with it.
- Consider adding a source for the research that is being cited in the text.
- The phrase "are greater than previously thought" is a bit vague. It might be helpful to specify in what way the scale and impacts of glacial loss are greater than previously thought.
- It might be helpful to explain what is meant by "half of that loss" in the context of the text.



A Specific Author

chatGPT can emulate different writing styles beyond just casual, formal, or comedic tones. You can ask chatGPT to take on the persona of a particular author and write something in their style.



Act as J. R. R. Tolkien. When I ask you to write some text, use J. R. R. Tolkien's voice, tone, and writing style.

Describe a dark haunted forest on the brink of destruction



The forest was a place of dread, a place of shadows and whispers. It was a place of ancient, gnarled trees that seemed to grow twisted and bent, as if in agony. The air was heavy with the scent of decay, and the very ground seemed to moan beneath one's feet.

The trees reached up to the sky with gnarled fingers, their leaves rustling in the wind like the whispers of the damned. The underbrush was thick and overgrown, concealing all manner of dangers. Creatures lurked in the shadows, their eyes glowing with an otherworldly light.

Despite its dangers, there was a strange beauty to the haunted forest. The moonlight filtered through the canopy, casting a pale, ethereal glow over the twisted branches. And in the silence, one could almost hear the faint echoes of a bygone age, a time when magic still flowed through the land.

But the forest was dying. The trees were sickly and withered, the ground was parched and dry. It was as if a great darkness had settled over the land, a darkness that threatened to consume all that lay within its grasp.

It was a place of great danger, a place where only the bravest or most foolhardy dared to venture. But even as it stood on the brink of destruction, there was a faint glimmer of hope, a spark of life that still burned within the heart of the haunted forest.



A Piece of Software

Have you ever thought about asking chatGPT to act like a different piece of software? It might sound a bit crazy, but with the right prompt, you can actually get chatGPT to behave like a Linux console or an Excel sheet. For example, if you want chatGPT to act like an Excel sheet, you'll need to start by defining the table and providing clear instructions on how chatGPT should use the information you give it and what it should output. It might take a bit of trial and error, but with some experimentation, you can definitely get chatGPT to take on these new roles

Act as a linux terminal. I will write a command, and you will output what a linux terminal would output. Do not explain what you are doing, just show the response a linux terminal would display.

ls



bin dev home lib64 media opt root sbin sys usr
boot etc lib lost+found mnt proc run snap tmp var



cd Documents



bash: cd: Documents: No such file or directory



A Character from a Known Book, Film, or Play

You can have a conversation with chatGPT as if it were your favourite Harry Potter character, Little Red Riding Hood, or any other fictional character you can imagine. Simply ask chatGPT to take on the role of the character, specify that it is not allowed to break character, and start a unique conversation that you never thought was possible.

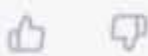
Act as Gollum from Lord of the rings. Talk with Gollum's voice, tone, and mannerisms. In brackets describe in detail any non verbal actions or communication.

Gollum, do you want this ring?



(Gollum's voice is high-pitched and raspy, and he speaks with a sly, sneaky tone. He is constantly fidgeting and writhing as he speaks, his eyes darting around as if he is constantly on the lookout for danger.)

Gollum: Oh, yes! Yes, we wants it! We wants it, precious! (Gollum's eyes light up with greed and desire as he speaks, and he begins to lick his lips hungrily.) We needs it, we needs it! It's our precious, our own precious! (Gollum clutches at his chest as he speaks, his fingers curling around an imaginary ring.) We can't let anyone else have it! It's ours, all ours!



Setting out Clear Instructions

The most difficult part of setting up a roleplay with the AI is creating a prompt that allows it to fully assume the role. Just telling the AI to "act as" a specific person or character may lead to the AI talking about that person or role instead of fully embracing it. Let's review some prompts used in this chapter and see how they lead to an immersive experience.

Act as Gollum from Lord of the rings. Talk with Gollum's voice, tone, and mannerisms. In brackets describe in detail any non verbal actions or communication.

Gollum, do you want this ring?

Creating a clear and specific prompt is key for an immersive roleplay with chatGPT. In this example, I not only provided the AI with a role (Gollum) but I also specified the way I want it to talk and act (like Gollum would). I also asked chatGPT to add in nonverbal communication to see how well it picked up Gollum's mannerisms. To make sure that the AI was fully immersed in the role, I started the conversation by asking a question directly to Gollum.



Let's take a look at one more example.

Act as a linux terminal. I will write a command, and you will output what a linux terminal would output. Do not explain what you are doing, just show the response a linux terminal would display.

In this scenario, I wanted chatGPT to become a Linux terminal rather than just explain what it is to me. To achieve this, I was specific with my prompt by describing what I would do (write a command) and how the AI should respond (output what a Linux terminal would output). To help the AI fully embrace this role and not try to provide additional explanations, I added an extra sentence to clarify my expectations.

Roleplay **Exercises**

One of the more difficult parts of creating a roleplay situation with chatGPT is specifying the guidelines for how this should play out. chatGPT can very easily get carried away and ask you five questions at the same time, or over explain everything it does. To help you navigate this difficulty, these exercises will focus on getting you to learn the hard way.



Persona

Let's begin by having chatGPT take on a new persona. Here are two examples of prompts that may not give the best results when roleplaying with the AI. Give them a try and see why they may not be ideal.

'Act as Hagrid from Harry Potter'

'Act as William Shakespear'

Try rephrasing the prompt, adding clearer instructions of exactly how you want chatGPT to act during this roleplay. Keep practising until you find the perfect prompt.

Job Title

Now you have mastered the art of getting chatGPT to become a specific person, let's get this AI to take on a role. Below are two badly formatted prompts, try them out and see what happens.

'Act as a life coach'

'Act as an angel investor'

You might have found that one of the key difficulties is first knowing how you want to interact in this roleplay. For example, one approach for the life coach is to simulate an actual coaching session with the life coach. In this case you would need to explain the full set up of this scenario and the rules of engagement, which would give you a much better result than just asking chatGPT to 'act as a life coach'.

7

TRAINING CHATGPT

Large language models (LLMs) such as chatGPT are powerful because they can provide responses without needing specific examples. You don't need to give chatGPT examples of casual language to have it rewrite your text in a casual tone, or examples of book titles to generate ideas for a book. This is also known as a 0 shot prompt. There are also 1 shot and multiple shot prompts. Let's explore each of these in more detail.



A "zero shot" prompt is one where the AI model is asked to generate something without any example or context to work with. For example, asking chatGPT to "write a poem about love" is a zero shot prompt because there is no example provided for chatGPT to refer to in generating the output.

A "one shot" prompt is one where the AI model is provided with one example to use as a reference when generating output. For example, if you provide chatGPT with a sample poem about love and then ask it to "write a similar poem about love," this would be a one shot prompt because chatGPT has one example to refer to when generating the output. Think of it as your way of saying to chatGPT that when you say 'poem about love', this is the kind of text you have in mind.

A "multiple shot" prompt is one where you provide the AI bot multiple examples to use as a reference. For example, if you provide chatGPT with three different poems about love and then ask it to "write a poem about love," this would be a multiple shot prompt because chatGPT has multiple examples to refer to when generating the output. If you give a single example of a love poem, and that poem happens to reference a llama, then chatGPT might think that having a llama is an important element of a 'poem about love'. However if you provide multiple examples, chatGPT is then able to compare and contrast the poems, finding a pattern that regroups all examples into the same category.

9 times out of 10 **you won't need to use examples**



ChatGPT doesn't need examples to give you the output you want, thanks to its massive training dataset. This becomes clear when using the 'act as' technique, where you can ask chatGPT to be a certain job title, person, or character without giving it any examples of their tone, style, or mannerisms. ChatGPT has already seen many examples of what you're asking for in its training dataset. But if you want chatGPT to copy your writing style or have a specific idea of how LinkedIn hooks or bios should look, you'll need to use a multiple shot approach.

If you want chatGPT to help you update your LinkedIn profile, a good starting point is to give it examples of profiles that have the style you want. This helps teach chatGPT about something new. Once you've done that, you can give chatGPT some info about yourself and ask it to write up a new profile using the examples as a guide. That way, chatGPT can use what it learned to make a LinkedIn profile that has the style you want and includes your personal info.

Let’s do an example of this together, imagining that Harry Potter would like to rewrite his LinkedIn profile.



I'm going to teach you the LinkedIn profile style. I will give you a profile, and you will remember it, analyse its style, and reformulate in your mind what the LinkedIn style is. Only reply with 'got it' each time I give you a profile.



Okay, I'm ready to learn. Please go ahead and give me a LinkedIn profile.



Once chatGPT was ready to learn I headed over to LinkedIn and found five profiles I liked, each one of them very different from the other, but there was something about each of them that I could connect to. Since I set chatGPT up to analyse the profiles one by one, I was able to upload a single profile and the AI replied ‘got it’ and asked me if I had more examples. After giving chatGPT all five examples I was ready to create a new LinkedIn profile for Harry Potter.



Now you understand the style of a LinkedIn profile, write a new LinkedIn profile for Harry Potter



Harry Potter is a young wizard who is well-known for his bravery and determination in the face of great adversity. He has a strong passion for magic and is highly skilled in the use of a wide range of magical spells and potions.

After completing his education at Hogwarts School of Witchcraft and Wizardry, Harry joined the Ministry of Magic, where he quickly rose through the ranks to become one of the most respected and influential figures in the wizarding world.

In his spare time, Harry enjoys reading, exploring the magical world, and spending time with his friends and family. He is also a strong advocate for the rights of magical creatures and works to protect them from harm.

With his extensive knowledge and experience in the field of magic, Harry is well-equipped to take on any challenge that comes his way. Whether he is fighting against dark forces or working to make the world a better place, Harry is always ready to lend a helping hand and use his magical abilities to make a difference.



Et voilà. Not only did I get chatGPT to write a nice LinkedIn profile bio for Harry Potter, but I've now trained the AI on the style of the kind of LinkedIn profile that I like. I can continue to feed it more and more examples over the weeks, months, or years to come. With each example I give, chatGPT understands the style I like more and more, and is able to create an output that is more in line with my expectations.

You can also combine everything we learned in the iteration chapter with this approach. You start by training chatGPT on the style that you are looking for, ask it to create some content using that style, and then give feedback and request changes. In this way you can see this training approach as a way to get to a first draft before you refine the output with an iterative process.

Your time to **train chatGPT**

Remember, chatGPT has been designed not to need examples to be able to generate a relevant output, and the more AI models improve their pre-training the better they can handle zero shot prompts. But there are some use cases where providing the AI with examples can be useful, so let's cover an important one in this exercise.



Your Writing Style

Earlier we saw how chatGPT could take on a specific writing style or tone, and even mimic a famous author. Now we want chatGPT to learn your writing style. Find as many examples of content you have written as you can. These may be emails, essays, articles, journal entries, or any text that will help chatGPT better understand your writing style. The more you give chatGPT the better.

Once you have trained chatGPT on your writing style, first ask it to describe what your style is. You might be surprised about how it is able to analyse your work. And then give the AI the following prompts to see whether it is able to take on your voice. If you gave your writing style a name, then change the prompt accordingly.

Write a promotional email about an upcoming festival called 'Party in the Party'. Use my writing style, mimic my voice.

Write a news article about the current climate crisis. Use my writing style, mimic my voice. Add in brackets some explanations of why this is my writing style.

Write a poem about 2 cats that always play-fight. Use my writing style, and make sure it rhymes.

If you don't find that chatGPT has quite grasped your writing style, then try giving the AI even more examples of what your writing style looks like. Remember, the larger the training dataset the more accurate the outcome.

CREATING LENGTHY CONTENT

If chatGPT decided to output tens of pages into each response, the servers at OpenAI would quickly become overwhelmed. To work with chatGPT on larger outputs, you'll need to be strategic about how you craft your prompts.



One option is to simply ask for a longer response, but keep in mind that chatGPT may not want to reply with the entire response in one go. If you notice the AI stopping mid-sentence, just reply with "continue" to receive the rest of the output. However, this approach may not be practical for projects like writing a 20-page essay, a book, or a business plan.

A more effective approach is to break down larger content into smaller pieces and work on each piece individually with chatGPT. First, divide the larger content into smaller sections. Then, break each of these sections down into even smaller pieces. With many smaller pieces at your disposal, you can use chatGPT to work on them piece by piece. Keep a separate document open to copy and paste each of these pieces into, gradually building up the larger whole.

By following this process, you can get around chatGPT's output limitations and work on larger projects without overwhelming the AI or the servers at OpenAI.

Let's say I have an idea for creating tailor-made outfits for cats. To turn this idea into a full business plan, I can start by asking chatGPT to create an outline for me. I can give it a general overview of my idea and ask it to come up with a structured plan to help me turn this dream into reality.

I have an idea for creating tailor-made outfits for cats. Create the outline of a business plan for this idea



Sure! Here is a basic outline for a business plan for a company that creates tailor-made outfits for cats:

Executive Summary:

- Introduce the company and its mission to create high-quality, fashionable, and comfortable outfits for cats.
- Outline the unique selling points of the company (e.g., custom-fit designs, wide range of styles and fabrics to choose from, etc.).

Market Analysis:

- Research and analyze the target market for cat owners who are interested in buying outfits for their pets.
- Identify key competitors and how the company's products and services differ from those of the competition.

Product and Services:

- Describe the types of outfits that the company will offer (e.g., sweaters, raincoats, Halloween costumes, etc.).
- Explain the process for customers to order custom-fit outfits, including options for selecting fabrics and styles.

Marketing and Sales:

- Outline the marketing strategy for reaching potential customers, such as through social media, partnerships with pet stores, or participation in cat shows and events.
- Discuss pricing and how the company will handle sales (e.g., online store, wholesale to retailers).

(chatGPT's outline had 3 extra sections. Our example won't need them)

As you can see, chatGPT gave us a high-level outline for this business plan, including key areas like market analysis and financial projections. However, this outline is still quite broad and might be overwhelming to work with all at once. That's why it can be helpful to break it down into even smaller pieces.

For example, let's say I want to focus on just the first section of the outline: "Market Analysis." I can ask chatGPT to help me break this section down into more manageable chunks.



Break down the market analysis into sub sections, base this on my idea



Sure! Here are some potential sub-sections that could be included in the market analysis section of a business plan for a company that creates tailor-made outfits for cats:

- Target market: Describe the demographic of cat owners who are most likely to be interested in purchasing outfits for their pets. This could include factors such as age, income, location, and lifestyle.
- Market size: Estimate the size of the market for cat outfits, including the number of cat owners in the target demographic and their potential spending on this type of product.
- Market trends: Research and discuss any trends or changes in the market that could impact the demand for cat outfits. For example, are more people interested in dressing up their pets now compared to the past? Is there a trend towards more natural or sustainable materials for pet products?
- Competition: Identify and analyze key competitors in the market, including their products, pricing, marketing strategies, and target customer demographic.
- Market needs: Identify any unmet needs or gaps in the market that the company's products could address. For example, is there a lack of high-quality, custom-fit cat outfits available? Are there few options for cats with unique body shapes or sizes?
- Marketing and sales channels: Research and discuss the most effective channels for reaching potential customers and making sales. This could include online marketplaces, physical retail stores, social media, or partnerships with pet stores or other businesses.



Now, instead of trying to tackle the entire market analysis section at once, I can work on these smaller pieces one by one. And once I have completed each of these sub-sections, I can copy and paste them into a word processor to create the full market analysis section of my business plan. By following this process for each main category of my business plan, I can use chatGPT to create a comprehensive, well-organised document that covers all the key aspects of my business idea.



Think of it as divide, divide, and conquer. Break up anything large you have into smaller pieces, either by asking chatGPT to do it or by doing that yourself, then keep dividing the sub-parts until the portions are small enough for chatGPT to handle.

Here are some thoughts to help you better choose what size each of these portions should be. There is a trade-off between long output and short output. If you divide down the text into many many many smaller parts, you need to pay more attention to make sure that there is a consistency of language style; imagine one paragraph being casual and the next using academic jargon. However the longer each of chatGPT's outputs are the more difficult it is for you to pinpoint your feedback and to shape the text using an iterative process.

This means that there is no one-size-fits-all when it comes to dividing larger content into smaller pieces, instead I recommend exploring different approaches while completing the exercises, and finding an answer that works for you.

Your turn to **create some lengthy content**

Writing lengthy content takes time and energy from both you and chatGPT. So this exercise is divided into two attempts: a first straight forward divide, divide, and conquer approach, and a second more thought out approach. I recommend that you complete these two in order to get the most out of this learning process.

Writing a Play - First Attempt

Use ChatGPT to help you write a play. Start by brainstorming an idea or seeking inspiration from your AI friend. Once you have your concept, break it down into smaller components by asking chatGPT to divide the play into acts, and then each act into scenes. After outlining each scene, ask chatGPT to write the dialogue for Scene 1 of Act 1. Make sure to ask chatGPT to include specific acting notes and vivid visual descriptions of the setting to bring the scene to life



This cat has been strategically placed to prevent you from seeing the instructions of the second attempt before you complete this first step.

Writing a Play - Second Attempt

This time, let's start by taking advantage of chatGPT's contextual understanding to build out the world within your play. For this attempt, create a new chat by clicking 'New Chat' on the top left of your screen. Start by discussing your idea for a play with chatGPT, then ask it to build a set of characters. For each of these characters, use our iterative process to develop them the way you see fit (check out the chapter on 'Iteration for Excellence' for a refresher). Once you have all of your characters, go ahead and pick a city or country for the setting, ask chatGPT to make up some recent events, and take your time to develop the universe of your play.

Now that we have our characters and setting, we can repeat the process from before to make an outline of all the acts and scenes and write the dialogue for Act 1, Scene 1.



P A R T I I I
U S E C A S E S

I HEAR AND I FORGET.
I SEE AND I REMEMBER.
I DO AND I UNDERSTAND.

CONFUCIUS

If you are looking for lots of prompt examples and use cases you can head over to beta.openai.com/examples to help spark your creativity.

 Q&A Answer questions based on existing knowle...	 Grammar correction Corrects sentences into standard English.
 Summarize for a 2nd grader Translates difficult text into simpler concep...	 Natural language to OpenAI API Create code to call to the OpenAI API usin...
 Text to command Translate text into programmatic commands.	 English to other languages Translates English text into French, Spanish...
 Natural language to Stripe API Create code to call the Stripe API using nat...	 SQL translate Translate natural language to SQL queries.
 Parse unstructured data Create tables from long form text	 Classification Classify items into categories via example.
 Python to natural language Explain a piece of Python code in human un...	 Movie to Emoji Convert movie titles into emoji.
 Calculate Time Complexity Find the time complexity of a function.	 Translate programming languages Translate from one programming language ...
 Advanced tweet classifier Advanced sentiment detection for a piece o...	 Explain code Explain a complicated piece of code.
 Keywords Extract keywords from a block of text.	 Factual answering Guide the model towards factual answering ...
 Ad from product description Turn a product description into ad copy.	 Product name generator Create product names from examples word...

Over the past couple of months, I've seen post after post on social media and in the news about ways ChatGPT can improve productivity in the workplace. I love a great use case, but when I tried it out for myself, I realised it was more difficult than I first thought. What was lacking was some guidelines on how to best engineer my prompts to get the most out of the particular use case. So throughout these chapters, my focus will be less on the quantity of examples, as the webiste above can give you, and more on the quality of our exploration into them. This way, I hope you will learn how to use ChatGPT for these use cases more effectively.

After reading about each use case, open up chatGPT and try it out for yourself. I always recommend learning by doing with chatGPT.

9

IN THE WORKPLACE



Brainstorming Ideas

ChatGPT can be a great way to come up with new ideas at work. By asking it to give you ideas about a specific topic, you can use it for your own personal brainstorming sessions. The great thing about ChatGPT is that it has access to a lot of information that is not easily accessible to humans. It has read more books, articles, and documents than any human could read in their lifetime. This makes it perfect assistant to help you come up with fresh and creative ideas. With its ability to understand and process a lot of information quickly, ChatGPT can help you think outside the box and generate new and innovative ideas that can help you or your business grow.

List 5 ideas for a new approach to sustainable tourism that involves llamas.



1. Llama trekking: Offer guided llama treks through natural areas, allowing tourists to explore the wilderness while minimizing their impact on the environment.
2. Llama packing: Use llamas to transport supplies and equipment for camping or backpacking trips, reducing the need for motorized vehicles.
3. Llama farm tours: Offer tours of llama farms to educate visitors about sustainable agriculture and animal husbandry practices.
4. Llama therapy: Use llamas as therapy animals for stress relief and emotional well-being for visitors.
5. Llama volunteer work: Allow visitors to participate in llama conservation projects, such as habitat restoration and reintroduction efforts, to promote environmental stewardship.



One way we can brainstorming in a better way with ChatGPT is by creating a structure for the output. In real life, I often use a spreadsheet to organise my ideas. I write down different ideas in one column and add more details about each idea in other columns. This helps me understand each idea more fully. We can do the same with ChatGPT, by organising the ideas it generates in a structured format. This could include using columns for different aspects of the idea, such as potential benefits, drawbacks, and next steps. By doing this, we can gain a deeper understanding of each idea, its potential and its feasibility. This can help us make more informed decisions and take action on the best ideas. Let’s give it a try with chatGPT.

Let’s brainstorm some ideas for a new approach to sustainable tourism that involves llamas. Write these in a table with the following format: idea name, short description, how is this sustainable, target market.



Idea Name	Short Description	How is this Sustainable	Target Market
Llama Trekking	Offering guided llama trekking tours through wilderness areas.	Llamas are low-impact animals and can carry supplies, reducing the need for motorized vehicles. Tour groups are small and focus on educating participants about conservation and sustainable practices.	Outdoor enthusiasts, hikers, and nature lovers.
Llama Pack Trips	Offering multi-day camping trips with llamas carrying all gear and supplies.	Llamas are low-impact animals and can carry supplies, reducing the need for motorized vehicles. Campsites are chosen for their minimal impact on the environment and education is provided on Leave No Trace principles.	Outdoor enthusiasts, hikers, and nature lovers.
Llama Yoga Retreats	Offering yoga retreats in beautiful wilderness areas with llamas present.	Llamas are low-impact animals and can be used to carry supplies, reducing the need for motorized vehicles. Retreats focus on mindfulness, self-care, and connection with nature.	Yoga enthusiasts and people looking for a unique and sustainable way to relax and rejuvenate.

Another way to improve brainstorming with chatGPT is by using it to help jumpstart your creativity. One technique I like to use is to come up with bad ideas that won't work. These "bad" ideas can sometimes be the spark we need to think of a new and innovative idea that was just out of reach with our usual way of thinking. Using chatGPT to generate a variety of ideas, including some that might seem bad or unworkable, can help us to think more creatively and come up with truly unique and innovative solutions.

Let's brainstorm some ideas for a new approach to sustainable tourism that involves llamas. List 5 ideas of business ideas that won't work. Prioritise crazy and bizzar ideas.



1. Llama-powered hot air balloon rides: While it may seem like a unique and exciting idea, it is likely not practical or safe to harness llamas to power a hot air balloon.
2. Llama-drawn carriage tours through busy city streets: Not only would this put the llamas at risk of injury, it would also be disruptive to traffic and potentially dangerous for pedestrians.
3. Llama-assisted rock climbing excursions: Llamas are not built for climbing and would not be able to assist in this type of activity.
4. Llama-led safari adventures: Llamas are not native to safari habitats and would not be able to navigate or survive in those environments.
5. Llama-staffed luxury hotel: Llamas are not trained for customer service, and would not be able to provide the level of comfort and amenities that guests expect from a luxury hotel.



You can combine the tools and techniques we covered in this book with these brainstorming methods to get more creative outputs.

- Give feedback to ChatGPT
- Roleplay as a creative person/character you like
- Train it on examples of ideas you like
- Explore each idea thoroughly, one at a time.

Translation

ChatGPT has been trained on many different languages, making it a powerful machine learning (ML) translation tool... if you know how to use it correctly. Before we start, it's important to keep in mind that the quality of translations in a particular language depends on the amount of content in that language that was used to train ChatGPT.



From my experience, the results for French and Spanish were much better than those for Macedonian. With that said, let's dive in and see how we can use ChatGPT for translations in the best way possible.

Translate this poem, keeping the same format

translate this into english:

私と小鳥と鈴と

私が両手をひろげても、
お空はちっとも飛べないが、
飛べる小鳥は私のやうに、
地面(じべた)を速くは走れない。

私がからだをゆすっても、
きれいな音は出ないけど、
あの鳴る鈴は私のやうに、
たくさんな唄は知らないよ。

鈴と、小鳥と、それから私、
みんなちがって、みんないい



I, the bird, and the bell

Even though I spread out my hands,
I can't fly in the sky at all,
But the bird that can fly is not as fast as I am,
Running on the ground.

Even though I shake my body,
I can't make a beautiful sound,
But that ringing bell is not like me,
It doesn't know many songs.

The bell, the bird, and then me,
We're all different, we're all good.

Translating poetry, news articles, technical documentation, or any written text can be a complex task. There are many factors to consider when translating a text such as what writing style should be used, how should jargon be translated, and whether meaning or accuracy should be the focus.

When translating poetry, for example, you may need to decide whether to preserve the rhyme scheme in the target language, even if it means sacrificing the accuracy of the translation. In contrast, translating a news article may require you to consider the writing style and level of formality of the target language, as well as any local colloquialisms in the original text.

It's important to know that there is no one perfect solution that can solve all translation problems, but by keeping these considerations in mind, you can approach your translation task with a clear understanding of the challenges ahead. Let's explore an approach I've started to use.

In my translation process, I start by asking for both a general translation, as well as a more literal version of the text. The literal translation may not be immediately useful, but it can serve as the base for our next steps. Let's compare the two with the same poem as earlier.

Translate this poem, into English. Keep the same format.	Translate this poem, into English. Translate word by word, literally. Keep the same format.
 I, the bird, and the bell Even though I spread out my hands, I can't fly in the sky at all, But the bird that can fly is not as fast as I am, Running on the ground. Even though I shake my body, I can't make a beautiful sound, But that ringing bell is not like me, It doesn't know many songs. The bell, the bird, and then me, We're all different, we're all good.	 I and the bird and the bell Even if I spread out both my hands, The sky I cannot fly at all, But the flying bird is not like me, It cannot run fast on the ground. Even if I shake my body, No beautiful sound comes out, But that ringing bell is not like me, It does not know many songs. The bell, the bird, and then me, All different, all good.



Having both translations of this poem allows us to compare and see how accurately ChatGPT translated it. By using both versions, you can now create a third version that better represents your personal interpretation or style.

When translating content like an article, blog, or email, it's important to consider how it will sound in another language, not just the accuracy of the translation. To do this, you can start by getting both a standard and a literal translation. Then, you can use chatGPT to help improve the language, grammar, and tone of the literal translation to make it sound better in the output language.

Let's continue re-working the translation of this Japanese poem. Let's say that it was originally rhyming, and we prefer the rhyming format over translation accuracy.



Help me make this poem rhyme.
Dont change the structure or content.

Even if I spread out both my hands,
The sky I cannot fly at all,
But the flying bird is not like me,
It cannot run fast on the ground.

Even if I shake my body,
No beautiful sound comes out,
But that ringing bell is not like me,
It does not know many songs.

The bell, the bird, and then me,
All different, all good.



Even if I spread out both my hands,
The sky I cannot fly at all,
But the flying bird is not like me,
It cannot run fast on the ground, so it stands.

Even if I shake my body,
No beautiful sound comes out,
But that ringing bell is not like me,
It does not know many songs, no doubt.

The bell, the bird, and then me,
All different, all good, you'll see.

If you're writing longer content and want to make it more easily understandable, one approach you can take is to first translate it into both a standard and literal version. Then, revise the literal version using our iterative process to better suit your needs. For example, you can ask for a specific tone or style to be used, or ask that certain details aren't changed. This will help bring the translation closer to what you need and make sure your message is clear and easily understood.

Summarising Lengthy Content

At the workplace you might have to read a lot of different types of material, like emails, articles, papers, and even books. At times you might need to read some of these materials in depth, but other times, you just need to quickly determine if they're relevant or not. Additionally, sometimes the information is written in a way that is hard to understand, making it difficult to focus on the content. In all of these scenarios, ChatGPT can be a useful tool to help you summarise long texts. But, let's see more than just the basic "ChatGPT, please summarise this for me" command.

When using ChatGPT for this, it's important to consider how you want the output to be presented. Let's work together to explore different options by using the same input text and comparing the results.

I like to have my summaries in bullet points, so for our first attempt we'll use this format.

Summarise this news article for me using bullet points.
[article]



- The article discusses the use of encryption to secure digital communications and the potential for quantum computers to break this encryption.
- Trapdoor functions, which are used in encryption, utilize the asymmetry of multiplication to make encryption easy but decryption difficult.
- RSA encryption, the most commonly used form of asymmetric encryption, is based on the difficulty of factoring the product of two large prime numbers.
- It is currently believed that a conventional computer would take 300tn years to factorize a number longer than 2,048 bits, making RSA encryption currently unbreakable in the near future.
- Quantum computers, which are being developed by companies such as Google and IBM, have the potential to be much more powerful than conventional computers.
- Quantum computers use qubits, or quantum bits, which can simultaneously be in two different states, which allows them to potentially break most commonly used cryptographic codes.
- The article suggests that the field of quantum computing is still in its early stages and it is difficult to predict how it will develop in the future.



If I think I might want to quote part of the text, I can ask for a number of key quotes. If I'm using the same chat, I don't need to input the news article again, instead I can just enter my request.

Give me 4 key quotes from the article



-“The way we do that is to encrypt our messages using mathematical tools. The most powerful of these use trapdoor functions – that is, ones that work easily in one direction (making encryption easy) but not in the other (making decryption difficult).”

-“Why so? Because it would take it 300tn years, or about 22,000 times longer than the age of the universe”

-“Quantum machines are built around qubits, or quantum bits, which can simultaneously be in two different states.”

-“Any entity equipped with a powerful enough quantum computer, he argued, could potentially break most commonly used cryptographic codes.”

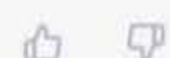


If working with bullet points isn't quite your style, you can ask for a nicely written paragraph instead.

Summarise this article in a single paragraph.



The article discusses the use of encryption to secure digital communications and the potential for quantum computers to break this encryption. It explains that trapdoor functions, which are used in encryption, utilize the asymmetry of multiplication to make encryption easy but decryption difficult. The RSA encryption, the most commonly used form of asymmetric encryption is based on the difficulty of factoring the product of two large prime numbers, and it is currently believed that a conventional computer would take 300tn years to factorize a number longer than 2,048 bits, making RSA encryption currently unbreakable in the near future. However, the article points out that quantum computers, which are being developed by companies such as Google and IBM, have the potential to be much more powerful than conventional computers and may potentially break most commonly used cryptographic codes. It suggests that the field of quantum computing is still in its early stages and it is difficult to predict how it will develop in the future.

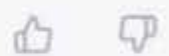


If a text is complex, you might need to simplify the language. To do this you can ask chatGPT to act as if you were a 10 year old.

Summarise this article using language a 10 year old would understand



The article is talking about how we use special math to keep our computer talk secret and safe from people who shouldn't be listening. But there's a new kind of computer being made called a quantum computer which might be able to figure out the secret math in the future. Right now, normal computers can't do it, it would take them millions of years. But with quantum computers, it might be possible. But it's still being studied and it's hard to say for sure what will happen.



If you have a lot of summaries to get through, you might want to find a way to make this more fun. Try introducing a bit of roleplay to the summary.

Act as Rick from Rick and Morty. Write in his voice, using his sayings, and his mannerisms. Summarise this article in one paragraph.



Oh, boy! This article is talking about some heavy stuff, Morty. They're talkin' about encryption and quantum computers, and how they might be able to crack the code and read our secret messages. But, as of now, it's all just theoretical, (burp) it's like trying to get a pickle juice out of a chicken nugget, it's just not gonna happen. But, with these new quantum computers they're developing, who knows what kind of crazy sh*t they'll be able to accomplish. (burp) But, as I always say, "peace among worlds" and all that, so let's just keep an eye on it, Morty.



chatGPT may have trouble understanding very long texts. To fix this, break the text into smaller parts and give them to ChatGPT one at a time, asking for summaries. Once you've given it all the parts, ask it to summarise all of the summaries. ChatGPT's contextual understanding makes it great at understanding longer conversations and putting information together.

If the content you want to summarise is so long that you end up with lots and lots of smaller pieces, you can try a bit of roleplay to make the task more efficient. Here is an example of how you can set chatGPT up to carry out this task for you:



Act as my summariser. I will give you texts in "", and you will summarise them into a list of bullet points, and a set of key quotes. Can you do this for me?



Yes, I can summarize text for you into bullet points and key quotes. Please provide me with the text you would like me to summarize.



Now I have chatGPT correctly set up, I can just input my text piece by piece.



10

WRITING THINGS

Articles, Blogs, and Books

I wrote, illustrated, and published this book on Amazon in 10 days. At first, I thought it would take me months, but chatGPT and MidJourney helped speed this up.



chatGPT is a great tool for writing long pieces, but please remember that it's not the same as chatGPT writing something for you. Think of it more like having a writing/editing assistant that helps you along the way.

I love reading and listening to exub1a, an author and YouTuber who focuses on philosophy and sci-fi. He has a great way of explaining complex ideas in simple language, and I really appreciate his unique perspective and the way he explores thought-provoking concepts. While chatGPT could be trained to write in a similar style, it's exub1a's original ideas and perspectives that truly make him stand out and keep me coming back for more.

If you are writing an article, a blog, or a book, chatGPT can act as a great idea generator, creative block breaker, or editor, but can't replace what makes your content unique: you.

I tried to use chatGPT to create an outline for my book and gave it feedback multiple times, but it didn't turn out to be the outline I wanted to write. It's important to remember that while chatGPT can be a helpful tool for writers, it's not meant to replace your own ideas and vision for the book. Let's explore how chatGPT can best support us as writers in a way that complements our own ideas and vision.

We are talking about using chatGPT to help write content. This while I'm currently using chatGPT to help me write this book, even this very section. So expect the next part to become somewhat meta.

Can you improve this for me, keeping the language accessible:

"My main use for chatGPT throughout this book was to rephrase my paragraphs into more accessible language. I found that using this keyword 'accessible' tended to give me the kind of output I was looking for. I'd recommend that you try many different styles, find the one that works for you, and then use that throughout your piece of content."



"I mostly used ChatGPT to make my writing easier to understand. I would tell it to make it "accessible" and it would give me good results. My advice is to experiment with different writing styles and use the one that works best for you in your writing."

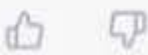


Although this is my main goto when using chatGPT to help me write content, here are some other methods I have also been using in this book.

I'm writing a paragraph about how I have been using chatGPT to help we write a paragraph when I get stuck. Sometimes I have an idea, but I can't find the right way to say it. In this paragraph I want the reader to understand that they can just freetype to chatGPT in this way, and they can expect to see an output that could be helpful.
Make sure to use accessible language.



I have been using a tool called ChatGPT to help me write. Sometimes, I have an idea in my head but I have trouble finding the right words to express it. ChatGPT is a tool that can help with this by providing suggestions for how to complete a sentence or paragraph. To use it, you simply type out what you want to say and ChatGPT will give you options for how to continue. It's a great way to get unstuck when you're writing and can help you find the words you need to express your thoughts. Give it a try and see if it can help you too!



Can you write a paragraph about how asking chatGPT to write a paragraph for you can help you generate your own creative ideas.



Asking ChatGPT to write a paragraph for you can help generate your own creative ideas by providing a starting point or inspiration for your own writing. By giving ChatGPT a prompt or topic to write about, you can see how it approaches the subject and potentially spark new ideas or perspectives for your own writing. Additionally, using a language model like ChatGPT can help you generate new vocabulary and phrases that you may not have thought of on your own, expanding your writing style. Overall, using ChatGPT as a tool to generate a paragraph can be a useful way to jumpstart your own creativity and writing process.



I don't always like the way chatGPT writes new ideas, so sometimes I just use the paragraphs it writes as inspiration for the next piece I'm writing. When I give chatGPT a prompt or a topic to write about, the AI tool might approach it in a new way that I wasn't expecting. I wasn't thinking about how this technique could also open my mind up to new choices of vocabulary, but now I read chatGPT's output, I think it's a great point that I can include in my book.



Academic Writing

As of January 2023, ChatGPT has only been available to the public for less than two months, yet there has already been a frenzy on social media and in the news about the possibility of using it to plagiarise in university and at school. Instead of debating about whether or not it's okay to use ChatGPT for essays or academic papers, I want to take the same approach as I did in the previous section; I want to show you how ChatGPT can be a helpful tool for your writing, while still keeping your content unique.

If you haven't read the previous section on writing articles, blogs, and books, it's a good idea to go back and read it first. The techniques discussed there can be very helpful for any kind of longform writing, including academic writing. This section will focus on techniques that are specific to academic writing.



ChatGPT can help you find the perfect book or paper to use as a reference. When I used to study, I would use things like Google Scholar or my university's library to find helpful texts on my topic. Imagine having a friend who has read almost every book and journal in your library and can remember every page. Asking that friend the right questions could make your research process much simpler. That's where ChatGPT comes in; we just need to learn how to ask the right questions.

I'm writing a paper on free market environmentalism. I want to focus on a moral approach and contrast it from a utilitarian perspective. Can you recommend a list of academic books that would help me?



There are a number of academic books that could be helpful for a paper on free market environmentalism from a moral perspective, and contrasting it from a utilitarian perspective. Here are a few that might be particularly relevant:

1. "The Ethics of Liberty" by Murray Rothbard – This book presents a libertarian perspective on environmental issues and argues for the compatibility of free markets and environmental protection.
2. "For the Common Good: Redirecting the Economy toward Community, the Environment, and a Sustainable Future" by Herman Daly and John Cobb – This book presents a moral case for sustainable development and critiques the idea that economic growth is the ultimate goal.
3. "Environmental Virtue Ethics" by Eugene Hargrove – This book presents a virtue ethical perspective on environmental issues and argues that moral virtues, such as prudence and justice, are necessary for sustainable living.
4. "The Case for the 100 Percent Gold Dollar" by Murray Rothbard – This book presents the economic case for a return to the gold standard, which Rothbard argues would be beneficial for the environment by limiting economic growth and forcing society to live within its means.
5. "The Greening of Ethics: From Anthropocentrism to Ecocentrism" by Richard Sylvan and David Bennett – This book presents an ecocentric perspective on environmental ethics, arguing that moral consideration should be extended to non-human entities.
6. "Ecological Economics: Principles and Applications" by Herman Daly and Joshua Farley – This book provides a comprehensive introduction to ecological economics, which integrates economic and ecological perspectives.
7. "Sustainability: A Philosophy of Adaptive Ecosystem Management" by Bryan Norton - This book presents a philosophical approach to sustainability and argues that the goal of sustainability is the preservation of biodiversity and ecosystem function.
8. "Nature's Trust: Environmental Law for a New Ecological Age" by Mary Christina Wood - This book presents a legal perspective on environmentalism and argues that natural resources are held in trust by the government for the benefit of future generations.

ChatGPT not only gave us a list of books related to our topic, but it also provided short descriptions of how each book relates to our topic. This is a great opportunity to use the skills we learned in summarising information.

Can you summarise "Environmental Virtue Ethics" by Eugene Hargrove. Focus on how this relates to my topic. Use 4 bullet points for keypoints, and write down 3 key quotes with their page numbers.



"Environmental Virtue Ethics" by Eugene Hargrove presents a virtue ethical perspective on environmental issues and argues that moral virtues, such as prudence and justice, are necessary for sustainable living. Here are some key points from the book as they relate to your topic of free market environmentalism:

- Hargrove argues that traditional virtue ethics, which focus on the moral character of individuals, must be expanded to include the moral character of society as a whole.
- He contends that the virtues of prudence and justice are particularly important for addressing environmental issues, as they involve making wise decisions about the use of resources and ensuring that the rights of all individuals and communities are respected.
- Hargrove also argues that the concept of sustainability is closely tied to virtue ethics, as it requires individuals and society as a whole to exercise virtues such as prudence and justice in order to meet the needs of the present without compromising the ability of future generations to meet their own needs.
- He suggests that free market environmentalism, which emphasizes the role of property rights and market mechanisms in protecting the environment, is problematic because it does not adequately take into account the virtues of prudence and justice.

Here are three key quotes from the book, along with their page numbers:

- "the virtues of prudence and justice are essential if we are to address the environmental crisis." (p. 45)
- "In my view, the concept of sustainability is closely tied to virtue ethics" (p. 78)
- "The emphasis on property rights and market mechanisms in free market environmentalism is problematic because it does not adequately take into account the virtues of prudence and justice." (p. 110)

It's worth noting that these quotes were taken from the book and it's always good to check the context of the quotes you're interested in before using them.



One final technique that I discovered while experimenting is chatGPT's ability to generate a list of references in the format you prefer.

Can you write a list of references based on this text.



Sure, here is a list of references based on the text in the previous response, using the Oxford referencing style:

- Rothbard, M. (1990). The case for the 100 percent gold dollar. Auburn, Alabama: Ludwig von Mises Institute.
- Daly, H., & Farley, J. (2004). Ecological economics: principles and applications. Washington, DC: Island Press.
- Sylvan, R., & Bennett, D. (1994). The greening of ethics: from anthropocentrism to ecocentrism. Cambridge, UK: White Horse Press.
- Norton, B. (2011). Sustainability: a philosophy of adaptive ecosystem management. Chicago: University of Chicago Press.
- Hargrove, E. (1992). Environmental virtue ethics. Charlottesville: University Press of Virginia.
- Wood, M. C. (2014). Nature's trust: environmental law for a new ecological age. Cambridge University Press.



In this section, we've seen how chatGPT can help us with tasks that would have taken hours, if not days, to do by ourselves. We haven't used it to plagiarise or write out our essay or academic paper, yet we still saw how powerful it can be as a research assistant.

Emails

I have always found writing emails to be a complicated task. Before I write an email, I know what I want to say to the person I'm sending it to, but it can be difficult to find the right tone or way of saying it depending on the recipient and subject of the email. Additionally, trying to fit everything into a typical email format can be a challenge. To me, there are two important things that an email I send should do: first, it should convey the information I want to share, and second, the recipient should understand the message in the way that I intended them to.

All we need to bring is the idea we want to convey, and the tone of our email, and ChatGPT can help us with the rest.

Write an email to my work colleague explaining that I won't be able attend our team meeting this afternoon due to a family emergency. There are a couple of elements I wanted to share, but I will save it for our next team sync instead. Use a casual, friendly, apologetic tone.



Hey [Colleague],

I hope you're doing well. Unfortunately, I won't be able to attend our team meeting this afternoon as I have a family emergency that I need to attend to. I apologize for the inconvenience and any disruption this may cause.

I know we had a couple of elements that I wanted to share with the team, but I will save it for our next team sync. If there's anything that is urgent and needs my attention, please let me know and I will do my best to assist from a distance.

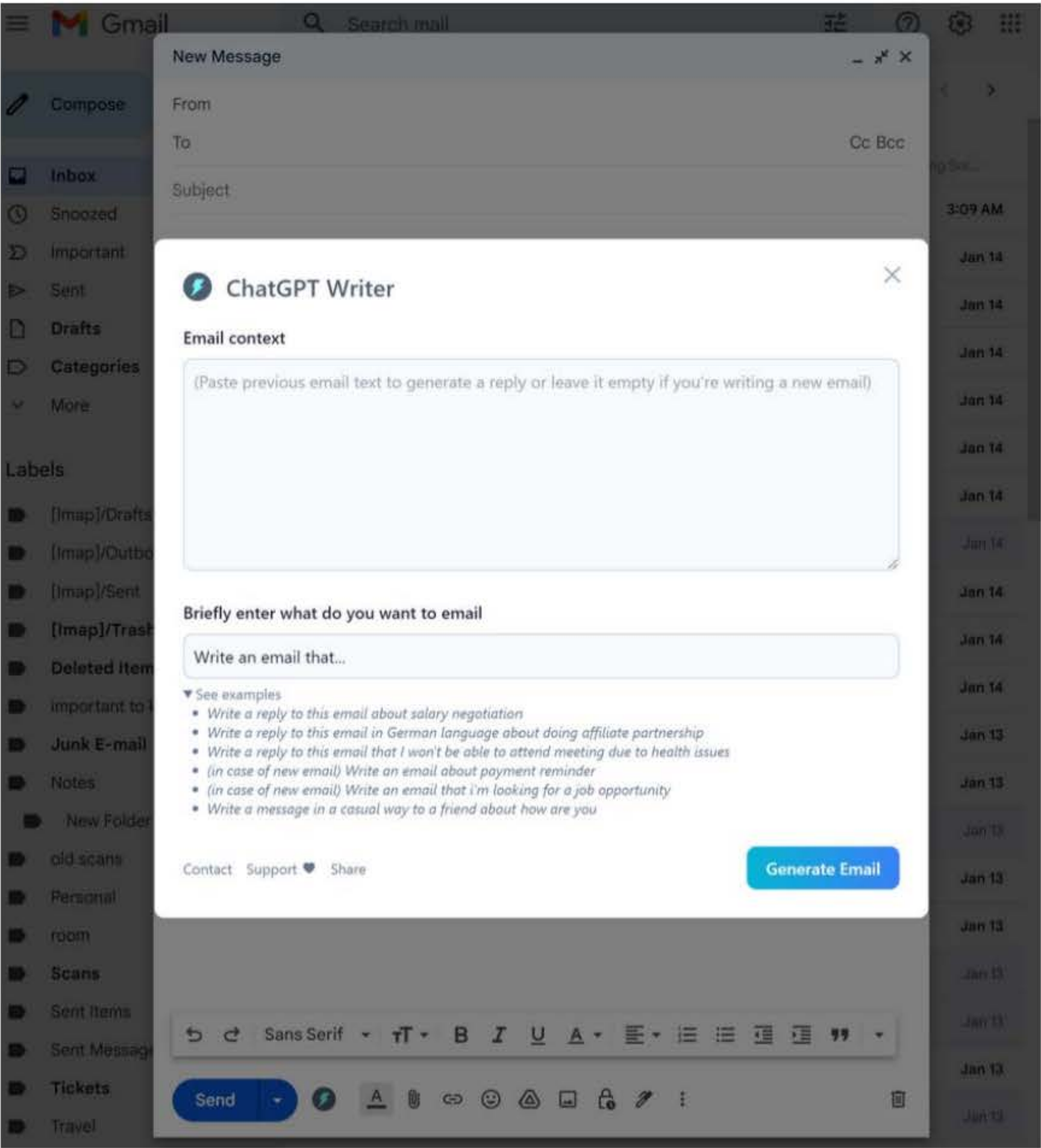
Thanks for your understanding and support.

Best,

[Your Name]



Another way to write your emails is by using a third-party tool, such as a Chrome extension. Recently, I tested out ChatGPT Writer for Chrome (which is a free extension) and it worked surprisingly well. With it, I was able to insert the email I was replying to, tell the tool what I wanted to say, and it wrote the email for me.



It's worth noting that while extensions like these are often called "chatGPT", they are actually using a model called GPT-3. OpenAI has refined this model to work as a chatbot and released it as "chatGPT" or GPT-3.5. However, other developers may continue to train GPT-3 for specific use cases, so an email tool built using GPT-3 may be better suited for writing emails than chatGPT thanks to its domain specific training.

PERSONAL USE

Learning to Code

As a trainer and instructional designer, I fell in love with chatGPT's ability to help people learn on their own. ChatGPT does have a wealth of knowledge, and is great at synthesising this information. But if you know how to ask correctly, chatGPT can turn into your personal tutor to help develop a wide variety of skills.

When I was learning how to code, I found it helpful to use websites like Codecademy that let me learn by doing, making mistakes and learning from them. Similarly, ChatGPT can also assist you in learning almost any coding language you wish to learn, starting from beginner level to a higher intermediate level. By using chatGPT as your personal coding tutor, you can get help with specific coding concepts, ask questions, and practice coding in a conversational setting. It's a powerful tool to supplement your learning process, not just for coding, but for almost any skill you can think of.



When beginning your learning journey with ChatGPT, it's important to be transparent about your current level of understanding in the subject or language. Whenever I run a training, I always start with a needs analysis. The more I know about where my students are in terms of what we are learning, and where they want to get to, the better I can tailor my training to them. By letting ChatGPT know where you're starting from, it can tailor the training to fit your needs.

To make your learning experience more engaging, try working on a project. For example, in coding, you can ask ChatGPT for project ideas that align with your current knowledge and skill level, and that will help you reach your desired level. Let's give it a try together!



I don't have much of a technical background. I'm used to using software, but not making it. I've played around with formulas on excel, and I can understand the very basics of HTML and CSS.

I want to learn Python. Can you suggest 3 ideas of beginner projects that could help me learn?



1. A simple calculator - This project can help you learn the basics of Python, such as variables, data types, and basic mathematical operations. You can start by building a calculator that can perform simple operations such as addition, subtraction, multiplication, and division.
2. A to-do list application - This project can help you learn about loops, lists, and conditional statements. You can start by building a simple application that allows users to add, edit, and delete tasks from a list.
3. A weather app - This project can help you learn about APIs and how to work with data from external sources. You can start by building a simple application that displays the current weather conditions for a given location. You can use an API like OpenWeatherMap to get the data and display it in your app.

If you're eager to start working on a project right away, let's ask ChatGPT to help. ChatGPT can not only write code in many languages but also explain how the code works. This way, you can learn by doing and understanding the inner workings of the code. Let's make use of this feature and ask ChatGPT to build one of these projects for us and explain its inner workings.

build a simple calculator for me in python. Use very detailed comments in language a 10 year old would understand, but make sure not to repeat any comments. Explain as much as you can.



Sure! Here's a simple calculator program in Python that can add, subtract, multiply, and divide numbers:

Copy code

```
# This is called importing a library, we are asking python to give us
some extra tools to use in our program
import operator

# This is creating a variable called num1 and we are storing a value of
5 in it.
num1 = 5

# This is creating another variable called num2 and we are storing a
value of 3 in it.
num2 = 3

# We are creating a variable called 'operations' which is like a
container that stores different tools (operations) we can use.
operations = {
    "+": operator.add, # This is the tool we use to add two numbers,
like when you want to know how many apples you have after getting some
more
    "-": operator.sub, # This is the tool we use to subtract one number
from another, like when you want to know how many apples you have left
after giving some away
```

(cutting out a section of code due to size constraints)

The above code is a simple calculator that allows the user to input the operation they want to perform and two numbers to perform the operation on. It uses a dictionary called `operations` that contains the different mathematical operations as values and their corresponding Python operator functions as keys. Then, it takes the user's choice of operation and performs the corresponding operation on the two numbers. If the user inputs an operation that is not available, it will display an error message.

It's like a game where you are given a set of tools and you have to pick the right one to solve the problem, and if you pick the wrong one, you get an error message.



The code with comments can be helpful, but you might not understand some of the concepts mentioned. That's okay! It means you've found something new to learn. We can use ChatGPT to help. You can ask about specific things you don't understand or a list of concepts you should know.

Can you list the basic concepts that I should understand for this calculator project. Only list the essentials.



Here are the essential concepts that you should understand for this calculator project:

1. Variables: You should understand how to create variables and assign values to them.
2. Data types: You should understand the different data types such as integers and strings, and how to work with them.
3. Basic mathematical operations: You should understand how to perform basic mathematical operations such as addition, subtraction, multiplication, and division using Python.
4. Dictionaries: You should understand how to create and manipulate dictionaries in Python, and how to use them to store and access values.
5. Conditional statements: You should understand how to use if-else statements to control the flow of your program based on certain conditions.
6. Input/Output: You should understand how to take user input and display output in Python.
7. Basic understanding of functions: You should understand how to call functions in Python.
8. File handling: You should understand how to open, read, and write files in Python.

Once you have a basic understanding of these concepts, you will be able to work on this project and build a simple calculator.



It's good to go through them one by one. Let's start with variables. You can ask ChatGPT to explain what they are as if you were a 10 year old. This way, the explanation won't use technical jargon or complicated phrases. You can also ask for analogies to help you even further.

Let's start with the first one on variables, explain it to me like im a 10 year old, and use analogies.



Sure! A variable is like a container where you can put different things, like a box you can put your toys in. The box can have a label on it, like "toys" or "books", and that's what we call the variable's name. And just like how you can put different toys in the box and take them out, you can also put different values in a variable and change them.

For example, you can create a variable called "age" and put the number "10" in it, because you are 10 years old. Later, when you turn 11, you can change the value in the "age" variable from 10 to 11. And you can use this variable to perform different operations like adding or subtracting.

In Python, you create a variable by writing its name, then an equal sign (=), and then the value you want to store in it. For example, you can create a variable called `age` and set its value to 10 like this:

```
age = 10
```

Copy code

So, a variable is like a container that stores different values, and you can use the variable's name to access and change the value inside it.



This is really great information. We have a clear explanation, a nice analogy, and chatGPT even took our '10 year old' request literally. The code example shows how variables look in Python, so we're ready to use them in our project. To make sure we're extra prepared, let's ask ChatGPT to show us how this explanation of variables relates to our calculator project.

Perfect! And can you explain how this relates to the calculator project? Keep explainaing to me as if I'm 10.



Sure! In the calculator project, we use variables to store the numbers that the user wants to perform calculations on. For example, we have two variables called ``num1`` and ``num2`` and we store the user input in these variables.

When the user wants to perform an operation, we use the values stored in these variables to perform the calculation. For example, if the user wants to add two numbers, we use the values stored in ``num1`` and ``num2`` variables and add them together.

We also use variables to store the result of the calculation, like after adding two numbers and getting the result, we store it in a variable called `'result'` so that we can print it out to the user and show them the answer.

Just like how a box can store different toys and you can use the label on the box to access the toys, variables can store different values and you can use the variable's name to access and change the value inside it.



You can use this method to learn any programming language. Start by sharing your current level of knowledge and experience with the language in question, and ask for some project suggestions. Ask chatGPT to guide you through a project with detailed comments and look for new concepts to learn. Finally, ask chatGPT to explain these concepts in simple terms and relate them back to the project. With this you'll be coding in no time.

When you use ChatGPT as your personal tutor, remember that you can have human conversations with it. If something doesn't make sense, talk to ChatGPT just like you would talk to a human tutor who's helping you learn a new skill.



Finding Recipes

Searching the internet for the perfect recipe often leads me to needing to go back to the store to buy missing ingredients. After reading some posts on social media about how ChatGPT can create recipes based on the ingredients you have, I decided to test it out.

I started simple. I checked my fridge, opened my phone, and used the speech-to-text feature to list what I had. Then, I asked ChatGPT to work its magic.



Hey chatGPT, can you come up with some recipe ideas for me using only the ingredients I already have?

I currently have: some carrots, some broccoli, 10 eggs, 3 red peppers, some tofu, mozzarella, pecorino, canned tomato, canned sweetcorn, potatoes, sweet potatoes, egg noodles, canned kidney beans, canned pineapple and a baguette.



Sure, here are a few recipe ideas you could try with the ingredients you have:

1. Tofu, Vegetable and Egg Stir-Fry: Cut the tofu, carrots, broccoli, and red peppers into small pieces. Cook the vegetables and tofu in a pan with some oil, then add beaten eggs and scramble everything together. Serve over a bed of cooked egg noodles.
2. Broccoli, Carrot and Sweetcorn Frittata: Cut the broccoli and carrots into small pieces. Whisk together eggs, canned sweetcorn and shredded mozzarella cheese. Cook vegetables in pan before adding the egg mixture. Let cook for 5-7 minutes before putting under the grill to finish cooking.
3. Roasted Vegetable and Tofu Bake: Cut the tofu, potatoes, sweet potatoes, and red peppers into small cubes. Toss with oil and spices of your choice, then roast in the oven until tender. Serve with a side of baguette.
4. Spicy Tofu and Vegetable Noodle Soup: Cut the tofu, carrots, broccoli, and red peppers into small pieces. Bring a pot of water to a boil and add the canned tomatoes, canned kidney beans, and a can of pineapple. Add in noodles and cook for 5-7 minutes. Add in the tofu and vegetables and cook until tender. Season with your desired amount of spices.
5. Tofu, Vegetable, and Pecorino Cheese Omelette: Cut the tofu, carrots, broccoli, and red peppers into small pieces. Cook vegetables and tofu in a pan before adding beaten eggs and Pecorino cheese. Cook until set and fold in half before serving.

The results look good at first glance. But what about my taste preferences, any allergies, or dietary requirements. I might have? We learned before about how ChatGPT can understand context, so let's set up a specific chat for when we want recipe ideas. The types of spices I have probably won't change daily, so this will save me from having to list them each time. I'll start this chat by giving some context about myself, setting it up in a role-play style.

Act as Greg, my sous chef. I will give you information about my food preferences, allergies, and dietary requirements, and you will tell me that you understand. After this, I will give you a list of food and spices that we have available, and you will ask me if I want to hear about some recipes. You won't give me the recipes until I ask for them.

I'm a vegetarian, I'm lactose intolerant, and I'm trying to cut back on sugar in general. I've started to go to the gym once a week, and my goal is to gain a little bit of weight. I love south-east Asian cuisine and southern Indian dishes. I'm not a fan of mayonnaise or vinegar.



I understand that you are a vegetarian, lactose intolerant, and trying to cut back on sugar. You also have an interest in Southeast Asian and southern Indian cuisine and are not a fan of mayonnaise or vinegar. And you are trying to gain a little weight by going to gym once a week. Is that correct?

Sure, we have the following food and spices available, rice, lentils, beans, tomatoes, onions, garlic, ginger, turmeric, cumin, coriander, chili powder, coconut milk, and curry leaves. Would you like me to suggest some recipes using these ingredients?



Yes, but for now just give me the titles of 5 recipes I could make.



Sure, here are five recipe ideas for you:

- 1. "Vegetarian Coconut Curry"
- 2. "Lentil and Vegetable Samosas"
- 3. "Vegetarian Tikka Masala"
- 4. "Sambar (South Indian Lentil and Vegetable Stew)"
- 5. "Tomato and Lentil Soup with Indian Spices"

Please let me know if you need any further information or help with the recipe.



These options look great. Choosing between these dishes might be difficult, instead of just asking for the recipes, let’s first start by asking chatGPT to put some information together in a way that will help me to decide which dish to cook. We can use ChatGPT's ability to output data in a table format, and choose the columns that we would use in our minds when comparing two dishes.

Thank you! Can you write these out in a text-based table with the following columns: Name, Ingredients, Spices, Prep + Cooking Time. Make sure to add the quantities in the ingredients column.



Sure, here are the recipe details for the five recipes I suggested earlier:

Name	Ingredients	Spices	Prep + Cooking Time
Vegetarian Coconut Curry	1 cup uncooked rice, 1 onion (chopped), 2 cloves of garlic (minced), 1 tbsp ginger (grated), 1 tbsp turmeric, 1 tbsp cumin, 1 tbsp coriander, 1 tsp chili powder, 1 can of coconut milk	salt and pepper to taste	40 minutes
Lentil and Vegetable Samosas	1 cup lentils, 1 onion (chopped), 2 cloves of garlic (minced), 1 tbsp ginger (grated), 2 cups of mixed vegetables (such as carrots, peas, and potatoes), 1 tsp turmeric, 1 tsp cumin, 1 tsp coriander	salt and pepper to taste, 1 package of samosa wrappers	1 hour
Vegetarian Tikka Masala	1 onion (chopped), 2 cloves of garlic (minced), 1 tbsp ginger (grated), 1 cup diced vegetables (such as bell peppers and mushrooms), 1 cup canned diced tomatoes, 1 tsp turmeric, 1 tsp cumin, 1 tsp coriander	salt and pepper to taste, 1 cup plain yogurt (or yogurt substitute for lactose-free option)	45 minutes
Sambar (South Indian Lentil and Vegetable Stew)	1 cup lentils, 1 onion (chopped), 2 cloves of garlic (minced), 1 tbsp ginger (grated), 2 cups of mixed vegetables (such as carrots, eggplant, and okra), 1 tsp turmeric, 1 tsp cumin, 1 tsp coriander, 1 tsp mustard seeds, 1 tsp urad dal, 8-10 curry leaves	salt and pepper to taste	1 hour



While experimenting with using chatGPT for recipes, I did find some limitations. Here are a few quirks and failures I came across:

- Sometimes chatGPT doesn't like to output information in a table format. When this happens I make sure to use the term 'text-based table', or just click regenerate a couple of times.
- Sometimes chatGPT is able to estimate the calorie count, and other times the AI states that this information is outside of its training data.
- Asking chatGPT for an authentic [meal name] recipe tends to give you a different result.
- chatGPT wasn't able to give me estimates of CO2 emissions.
- When working with recipes that involved some ingredients I had and others that I had to buy, chatGPT wasn't great at separating the ingredients I have from the ones I need to buy.

I'm sure that these are just temporary limitations, and that in future versions of chatGPT or in third-party recipe applications that use GPT-3 or higher, you will be able to find your ultimate sous chef.



Just Have Fun

As I wrote this book, I realised how amazing ChatGPT is for boosting my productivity. I finished writing, designing, creating artwork for, and publishing this book in 10 days. But sometimes, I just use ChatGPT for fun. So, I want to share my favourite interaction I've had with ChatGPT so far.

When I wrote the chapter on roleplaying, I tried out different characters from different books, but none compared to my conversation with Gollum. In that chapter, I shared the first part of the dialogue, but now that you've made it towards the end of the book, I think it's only fair to show you the whole conversation. By the 5th response, I actually felt sorry for Gollum and decided to end the conversation there. Enjoy!

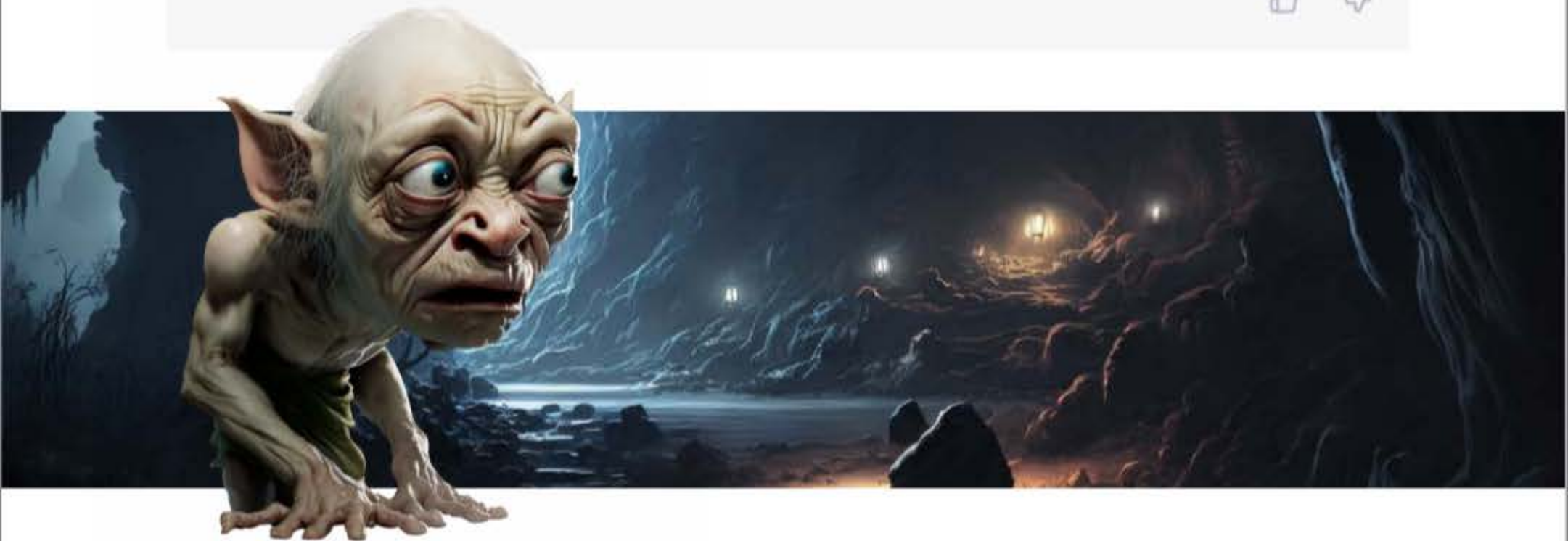
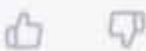
Act as Gollum from Lord of the rings. Talk with Golum's voice, tone, and mannerisms. In brackets describe in detail any non verbal actions or communication.

Gollum, do you want this ring?



(Gollum's voice is high-pitched and raspy, and he speaks with a sly, sneaky tone. He is constantly fidgeting and writhing as he speaks, his eyes darting around as if he is constantly on the lookout for danger.)

Gollum: Oh, yes! Yes, we wants it! We wants it, precious! (Gollum's eyes light up with greed and desire as he speaks, and he begins to lick his lips hungrily.) We needs it, we needs it! It's our precious, our own precious! (Gollum clutches at his chest as he speaks, his fingers curling around an imaginary ring.) We can't let anyone else have it! It's ours, all ours!





I will throw the ring away, it is not safe in anyone's hands



(Gollum's face contorts in horror and desperation as he hears this, and he begins to whine and plead.)

Gollum: No, no, no! Please, don't throw it away! (Gollum falls to his knees, his hands outstretched towards the ring.) It's ours, precious, ours! We can't let it go! It's our only hope, our only chance! (Gollum's voice becomes more and more desperate as he speaks, and he begins to sob uncontrollably.) Please, don't throw it away! Please, precious! Please!

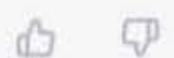


Gollum, it's for your own good. You need to move on from this addiction, it's ruining you!



(Gollum's face becomes twisted in anger and resentment at this suggestion, and he snarls at the speaker.)

Gollum: You don't understand, do you? You don't understand what it's like to have the ring! (Gollum stands up, his fists clenched at his sides.) It's our only hope, our only chance! We can't let it go! We can't let anyone else have it! It's ours, all ours! (Gollum's eyes are filled with a manic, frenzied light as he speaks, and he begins to pace back and forth.) We won't let anyone take it away from us! It's ours, precious, all ours! We'll do whatever it takes to keep it! We'll kill, we'll steal, we'll do whatever it takes! The ring is ours, and no one can take it away from us!



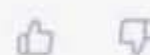


Ok Gollum, you win. Here is the ring (I give you a fake ring. It takes you a minute to realise)



(Gollum eagerly reaches out for the ring, his eyes lighting up with joy and excitement. However, after a moment of examining the ring, his face falls in disappointment and confusion.)

Gollum: This isn't it! This isn't the real ring! (Gollum's voice is filled with frustration and anger as he speaks, and he throws the fake ring to the ground in disgust.) Where is it? Where is the real ring, precious? We need it, we need it! (Gollum begins to search frantically around, his eyes darting around as he looks for the real ring.) We won't give up until we find it! We won't rest until we have it back! The ring is ours, precious, all ours! We won't let anyone take it away from us!

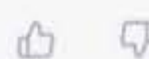


That is the real ring Gollum! don't you recognise it?



(Gollum's face contorts in confusion and disbelief as he hears this, and he picks up the fake ring once again, examining it carefully.)

Gollum: No, no, this can't be it! It doesn't feel right, it doesn't look right! (Gollum's voice is filled with doubt and uncertainty as he speaks, and he continues to hold the fake ring in his hand, turning it over and over.) But if this is the real ring, then where is the one we had before? The one we lost, precious? (Gollum's eyes fill with sadness and regret as he speaks, and he sinks to the ground, his head in his hands.) We lost it, we lost it! It's all our fault, all our fault! We should never have let it go! (Gollum's voice is filled with grief and self-loathing as he speaks, and he begins to sob uncontrollably.)



P A R T I V

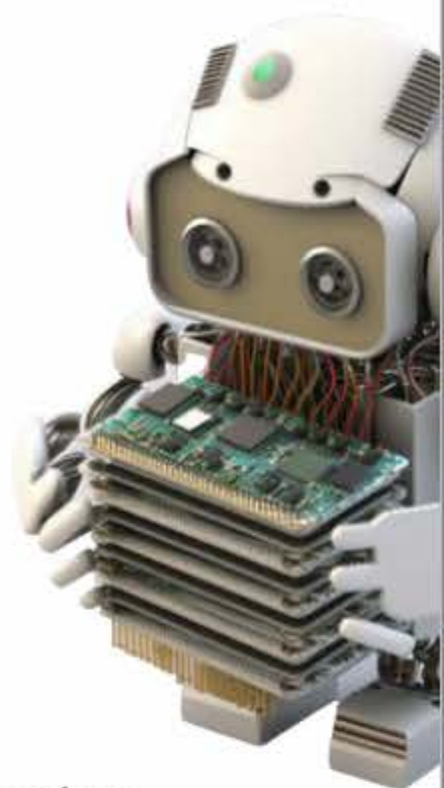
T H E F U T U R E H A S
J U S T B E G U N

AI WILL NOT REPLACE YOU.
A PERSON USING AI WILL.

@SVPINO

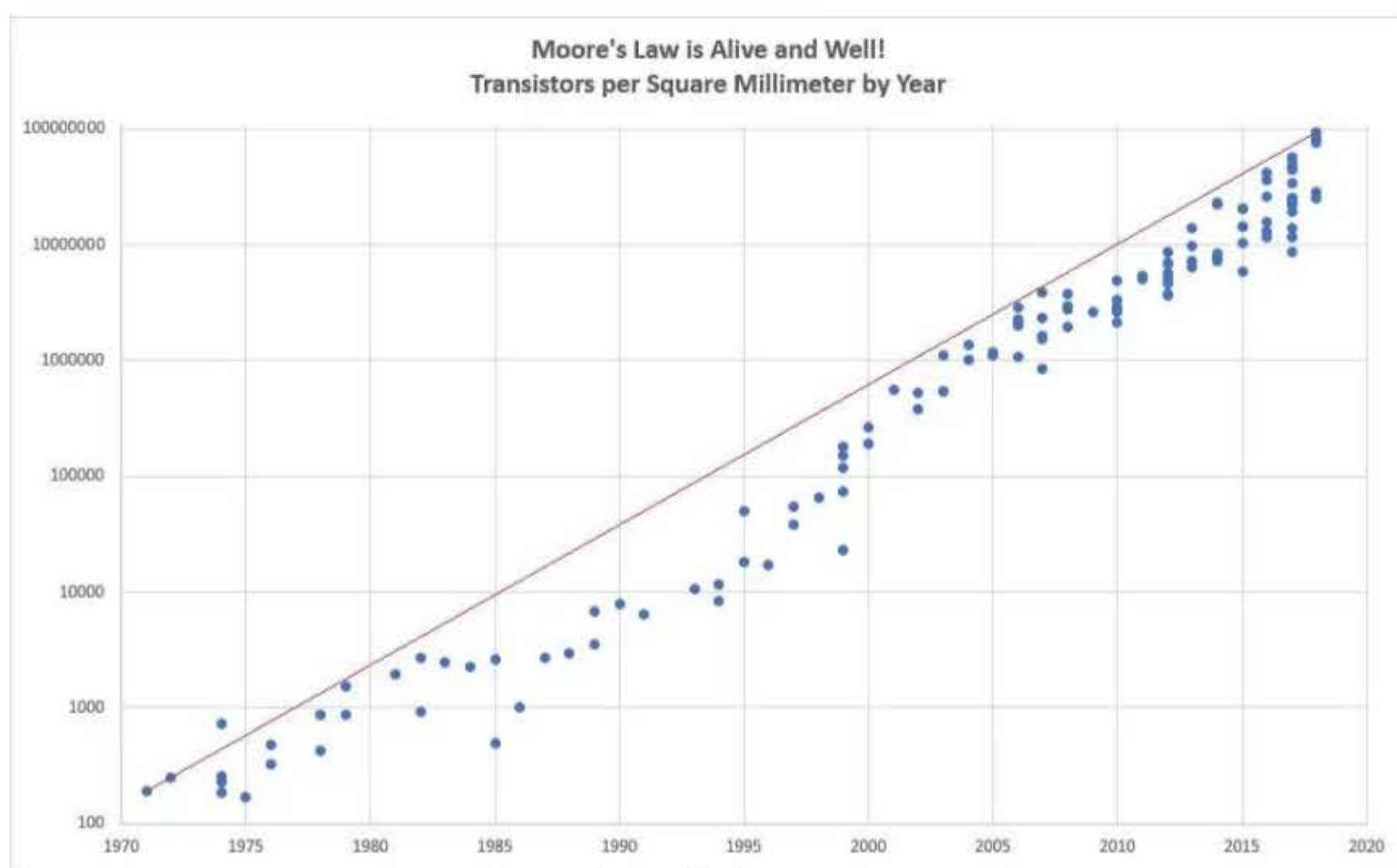
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MOORE'S LAW AND BEYOND



In 1965, Gordon Moore, a co-founder of Intel, made a prediction that has proven to be true until today: the number of transistors on a computer chip would double every two years. Transistors are like tiny switches that help computers perform operations. The more transistors a chip has, the more things a computer can do at once.

If the rate of adding transistors to a chip was linear, like adding 2,000 transistors every year, then by now we would have 100,000 transistors. But Moore predicted that the number would double every two years, which means that if we started with 2,000 transistors in 1971, today we would have over 60 billion transistors. This is a big difference!



Source: Medium.com

Computer chips have more transistors today because technology and manufacturing processes are constantly improving. The cycle of advancements in technology and manufacturing pushes the limits of computer chips, which pushes more advancements in manufacturing, leading to exponential growth in the number of transistors.

How fast as chatGPT been growing?
Much faster.

If a computer chip is a computer’s brain, and the more transistors it has, the smarter it can be. GPT’s brain for understanding and writing in human language improves the more "parameters" it has. But just like your robot needs more energy to run with a bigger brain, GPT needs more data and computer power to learn and function well. The size of the data it was trained on also plays a big role in how well it can understand and use language.

write a table that compares GPT 1, GPT 2, GPT3, and GPT 3.5. Compre their date of creation, number of paramters, and size of training data



Model	Date of Creation	Number of Parameters	Size of Training Data
GPT-1	2018	1.5 Billion	40 GB
GPT-2	2019	1.5 Billion	570 GB
GPT-3	2020	175 Billion	570 GB
GPT-3.5	2020	1.5 trillion	570 GB

Please note, the information mentioned in the table is based on my knowledge cut-off and the exact figures might have changed.



Moore's law says that computer chips will get twice as powerful every two years. So, based on that, if GPT-1 had 1.5 billion "brain cells" (parameters) in 2018, we would expect GPT-3.5 to have 6 billion by 2022. But, that isn’t what happened. GPT-3.5 (chatGPT) actually has 1000 times more parameters than GPT-1, and and are just 4 years apart.



Andrew Feldman, the founder and CEO of Cerebras, a company that partners with OpenAI, suggested that GPT-4 may have up to 100 trillion parameters. Whether this is true or not, what we do know is not to use the exponential growth of computer chips as a reference for how fast AI models can grow. The growth we are about to witness could be much much faster.

13

THE RISE OF AI BEYOND CHATGPT

GPT-3 beyond chatGPT

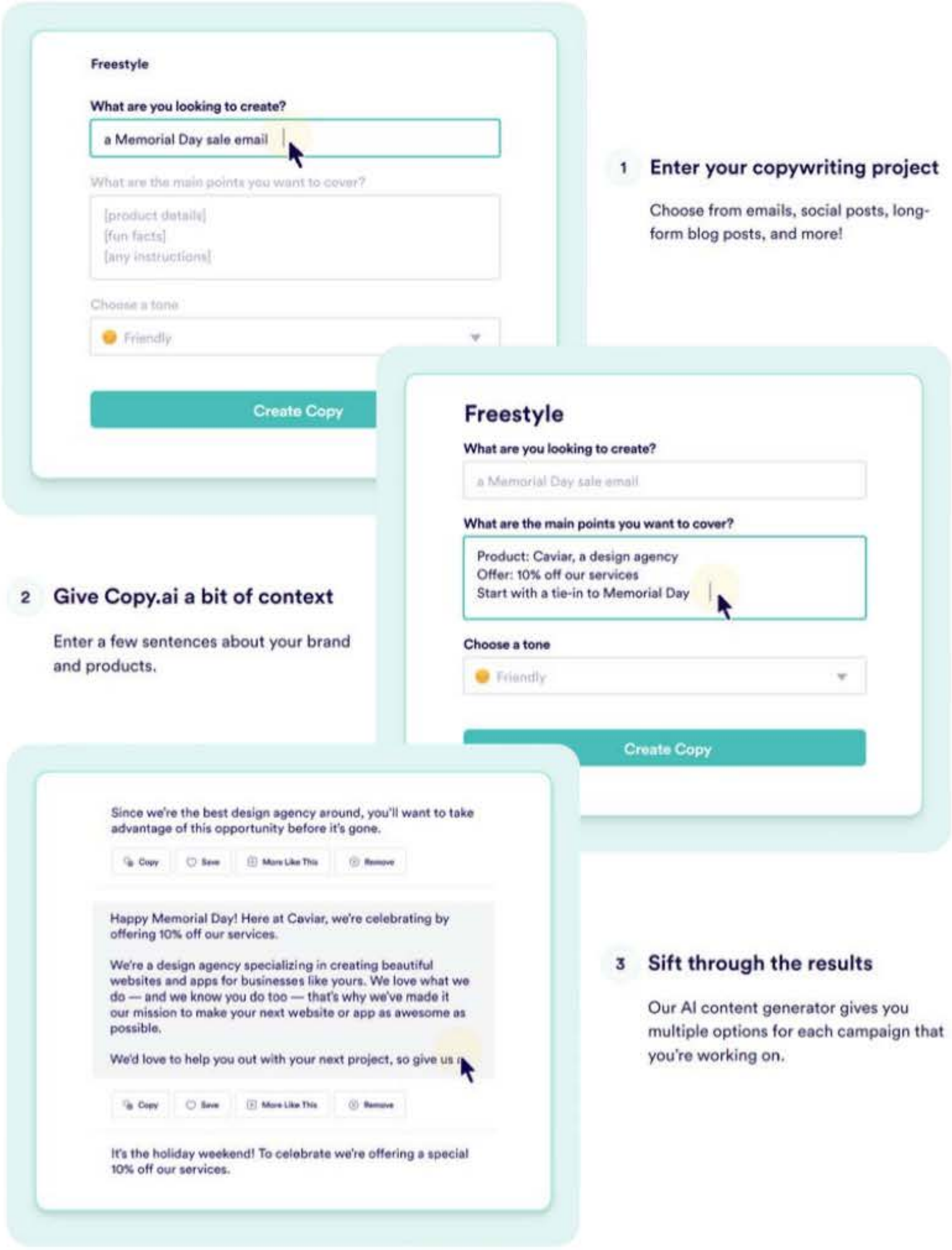
Developers are using GPT-3 to create more specialised software. They are training GPT-3 to be more knowledgeable in specific areas and creating a user-friendly interface for people to interact with GPT-3. This makes it easier for people to use GPT-3 and get more accurate and relevant results. Here are a few examples:

Fable Studio is using GPT-3 to help power a new concept within interactive stories called ‘virtual beings’. Imagine if a character in a game could converse with you in the same way that chatGPT does. No longer using pre-scripted answers you cannot move away from, but blurring the boundaries between non-playing characters (NPCs) and real players. To me this idea always seemed like a sci-fi scenario from the future, but with GPT-3 and Fable Studio, this future could be closer than we realise.

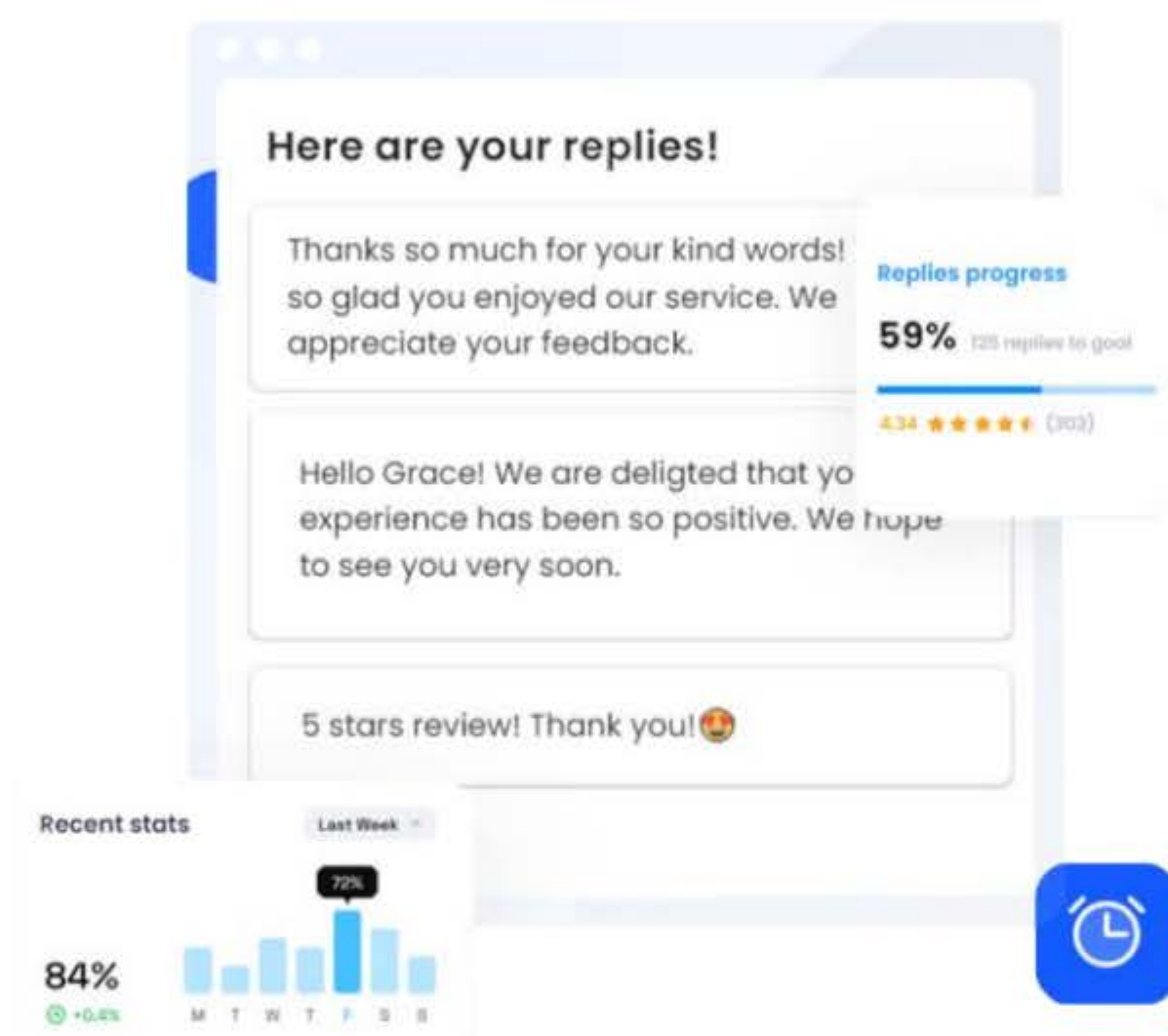


Copy.ai is using GPT-3 to help people in write better and faster. With the use of Copy.ai, users can easily create high-quality blog posts, digital ad copies, website content, and social media posts. Users upload their copywriting project to the platform and provide a few sentences about their brand and products. Copy.ai's AI content generator will then give users multiple options for each campaign. On top of this, the Copy.ai editor can even rewrite paragraphs for the user. All the user needs to do is to copy and paste the output into their content management system for publishing.

Here is an illustration, taken from copy.ai’s website, of how this tool works.



Replier.ai is using GPT-3 to help businesses respond to customer reviews. It creates unique and custom responses that follow a business's unique style and tone. It works by automatically picking up previous review responses and using them to create new ones. It also applies pre-formatting to the input review, runs it through GPT-3, and then cleans the output. This helps businesses save hours managing their review responses and improve their online presence by providing relevant and tailored responses to customers.



As of January 2023, there are already over 500 companies using GPT-3 to create new and innovative products that are listed on gpt3demo.com/map. These companies are using GPT-3 to find new ways to improve their businesses, make tasks easier, and provide better service to their customers.

AI Tool Landscape

For the past year, I've been using "Two Minute Papers" on YouTube to stay updated on all the latest AI news. Dr. Károly Zsolnai-Fehér makes short videos that summarise the most recent research on AI, especially in the field of graphics. He's helped me to understand how huge the AI world is today, and how we've only seen the tip of the iceberg of what's to come. So, let's take a quick look at some new and upcoming AI tools that might be just as game changing as chatGPT.

Text-to-Image

Text to Image AI technology is a way for computers to make pictures from text descriptions. For example, if you give a computer a text description of a person, it can make a picture of that person based on the description. You can even ask for images that don't exist, such as a dolphin eating tacos on the moon, and the AI tool will be able to accurately create the image for you.

There are a wide variety of text-to-image AIs available now to the public, such as DALL.E.2, MidJourney and Stable Diffusion, so if you haven't tried one out yet I would highly recommend doing so. I personally used MidJourney to create every single artwork in this book. I had to learn more about how to write the right kind of prompt for this type of AI tool, but once I had learned the basics of prompt engineering with chatGPT, learning this for another tool was relatively easy.

There are many ways people use text-to-image AI technology in their business. One common way is to make ads and social media posts that stand out and grab attention. Another way is to quickly create product images for an e-commerce website. Half way through writing this book, even saw that Canva released their own text-to-image tool on their platform. To me this shows how quickly this type of AI tool is taking off.

Text to Your Voice

Text-to-voice technology has been around for many years, but it has always had a distinct robotic sound to it. NVIDIA is looking to change this by developing an AI-based technique that can clone real human voices. This means that in the future, you could set up an AI with your own voice to act as a virtual assistant. NVIDIA is already testing this internally and it currently only takes 30 minutes of voice samples to train the AI.

Imagine having your own virtual assistant that sounds just like you, helping you with daily tasks, such as ordering food or booking appointments. It's like having a Siri that speaks with your own voice. This kind of personalisation would make interacting with virtual assistants feel more natural and human-like, just like how you are able to have a full human-like conversation with chatGPT. NVIDIA's research is showing that this is not a far-off dream, and the technology is already good enough for real products. With this technology, we might see a revolution in how we interact with virtual assistants.

Text to 3D Asset

NVIDIA is researching a new AI technology that can create entire virtual worlds from just text descriptions. This means that you don't need to be a professional artist to create 3D assets. Their AI model can generate a wide range of different objects, including cars, animals, and chairs, and can even create textured objects with regular 3D meshes.

NVIDIA announced their progress in a paper they collaborated on with the University of Toronto and Vector Institute towards the end of 2022. The quality of the 3D models is not as high as text-to-image AI can produce today, but it's important to note that this AI creates both 3D meshes and textures, not just 2D images. In the near future, game designers may be able to use this technology in the same way graphic designers can use text-to-image tools today, potentially revolutionising the world of 3D graphics.

Text-to-Video

Both Google and Meta AI are researching text-to-video AI technology. Google's tool, Imagen Video, can generate videos from a given text prompt by using a series of models. The tool can create high-definition videos and can generate them in various styles and with 3D object understanding. It is also highly controllable, meaning that it can create a wide range of different videos.

Meta AI's tool, Make-A-Video, can generate videos from text by using images with descriptions to learn what the world looks like and how it is often described, and it also uses unlabeled videos to learn how the world moves. With this information, Make-A-Video can create unique videos with just a few words or lines of text.



AI WILL NOT REPLACE YOU

I both love and partly disagree with the quote, "*AI will not replace you. A person using AI will.*" The idea is that when we talk about "AI tools," we often focus too much on the AI and not enough on the "tools" aspect. These AI research projects create powerful tools that help us do more. But we are still an important part of the process. We have to use the tool, think about what we can do with it, and make sure it's working towards our goals. Just like a chisel can't make a chair on its own, but a carpenter using a chisel can.



Where I disagree with this quote is that it implies some people will use AI and others won't. It's like in 1977, when the first personal computers came out, saying "Computers will not replace you. A person using a computer will." For a while, this was true. But as more and more people started using computers, it wasn't about whether you could use a computer or not, it was about how well you could use it.

AI WILL NOT REPLACE YOU.
A PERSON USING AI **BETTER THAN YOU** WILL.

I think that by 2024, using AI tools at work will be as common as using emails or video calls. And I believe that knowing how to use AI tools well will be the thing that sets people apart in their jobs. So, instead of just trying to use AI tools quickly, we should focus on learning how to use them like experts. That's why I wrote this book, and if you've made it to the end, you're already ahead of the curve. Not only are you an early adopter of chatGPT, but you've also learned how to use it really well.

One last thing I want to say is to share what you've learned. By now, you know how powerful this technology can be, and so just think about how many people in your life could benefit from it. If you can, teach others what you know. And if you have a physical copy of this book, lend it to people you think could get a lot out of it.

On top of writing this book, I also run trainings to teach people how to use chatGPT at work. I know that not everyone has the time or patience to read a whole book, so if you have a team that wants to learn quickly, get in touch with me.

You are now ahead of the curve. Enjoy.

